Block practice · Pathogens, immunity and public health

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Define pathogen, classify diseases and name the organism type causing each

Identify bacterial cell structures and compare with human cells

Describe how a named infection spreads, shows itself and is prevented

Explain antigen-antibody specificity and how monoclonal antibodies are made

Explain how vaccination or infection gives active immunity

Classify immunity as natural or artificial and active or passive

State what antibiotics treat and explain how resistance develops

Use aseptic technique to test antibacterial agents on agar

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Practice workbook · Define pathogen, classify diseases and name the organism type causing each

The term pathogen, communicable versus non-communicable conditions, and matching named diseases to virus, bacterium, protozoan or fungus.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-PATHOGEN-AND-NAME-CAUSE

1. Question 3 · part a · 2 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

.....

.....

.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

1. Question 3 · part a · 2 marks · Working and answer

Working and answer

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2. Question 4 · part b · 1 mark

June 2019 Paper 2

Further practice · Attempt part b only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

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number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

.....

.....

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2. Question 4 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

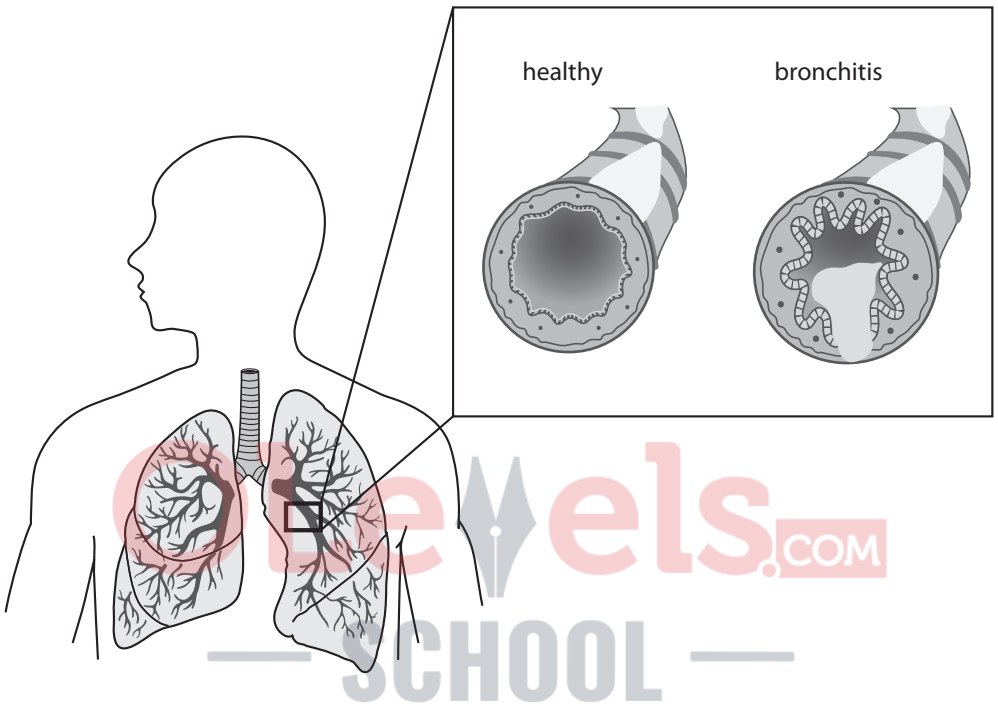
3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

6 Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.



(a) Name the part of the lungs affected by bronchitis.

(1)

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3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

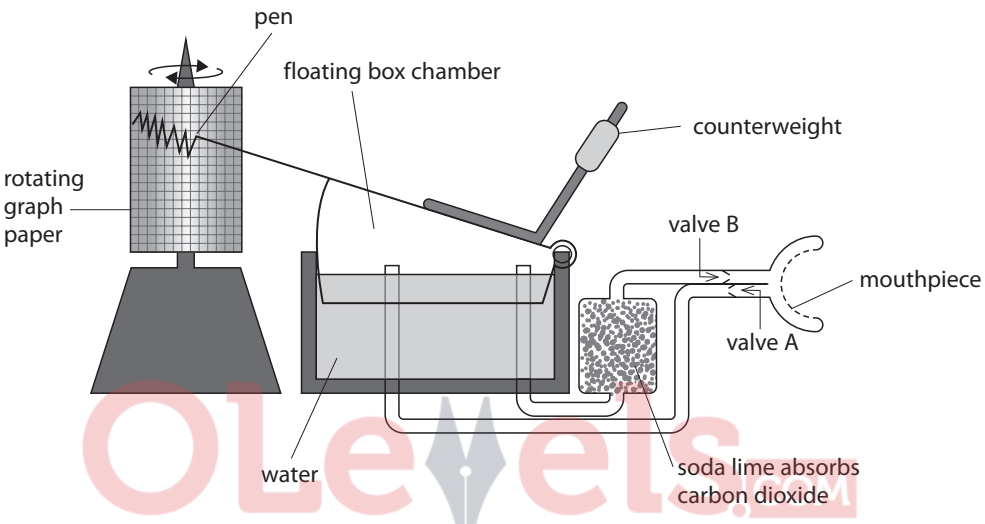
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(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.



(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

(4)

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3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

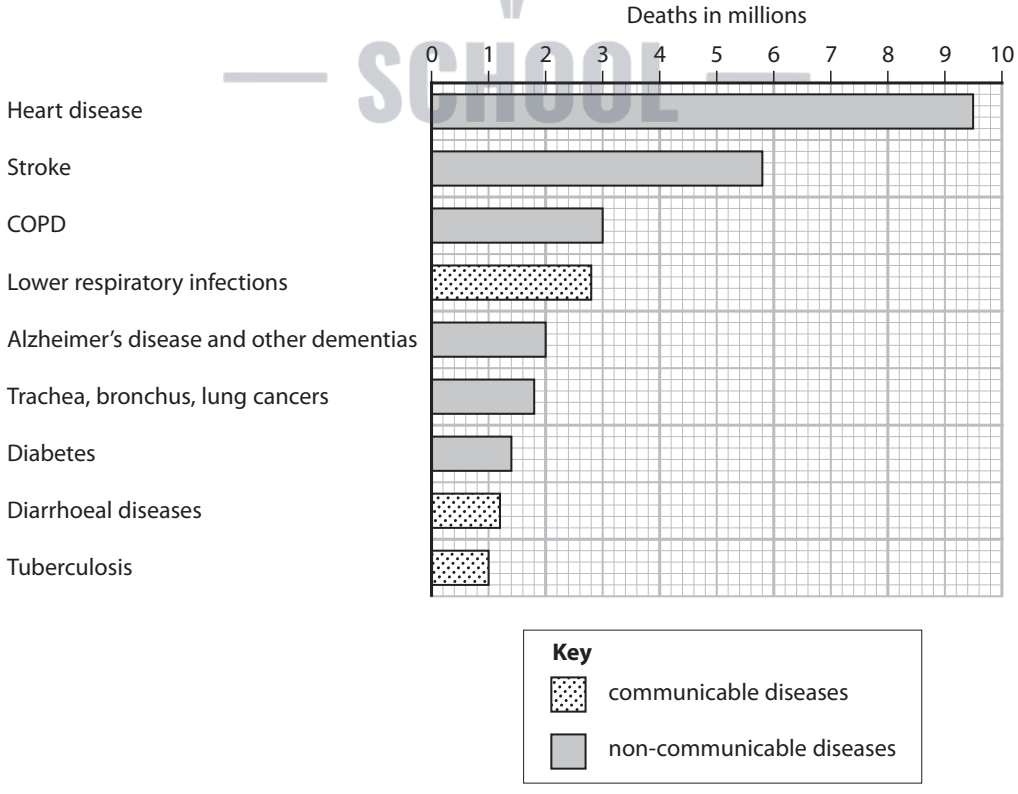
(ii) State one way to reduce risk to health when using a spirometer.

(1)

(iii) Explain why a person with bronchitis needs to take a longer breath to obtain a normal tidal volume.

(2)

(c) The bar chart shows the number of deaths in millions from different health conditions worldwide in 2016.



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3. Question 6 · part c.i · 1 mark

January 2020 Paper 1

Further practice · Attempt part c.i only.

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(i) More deaths are caused by non-communicable diseases than by communicable diseases.

Suggest what is meant by the term **non-communicable disease**.

(1)

.....

.....

.....

(ii) In 2000, the number of deaths from heart disease was 7.0 million.

Calculate the percentage change in the number of deaths from heart disease from 2000 to 2016.

(3)

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percentage change = ...%

(iii) Suggest a reason for the change in the number of deaths from heart disease from 2000 to 2016.

(1)

.....

.....

.....

(Total for Question 6 = 13 marks)

3. Question 6 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 7 · part a.i · 1 mark

November 2021 Paper 2

Further practice · Attempt part a.i only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease?

(1)

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics.

(2)

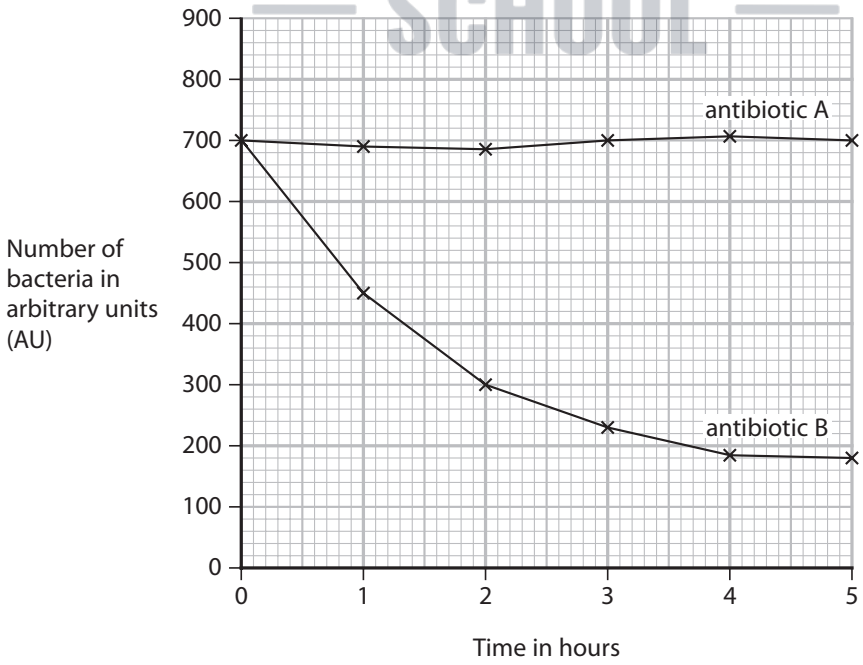
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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4. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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5. Question 8 · part a.ii · 1 mark

June 2023 Paper 1R

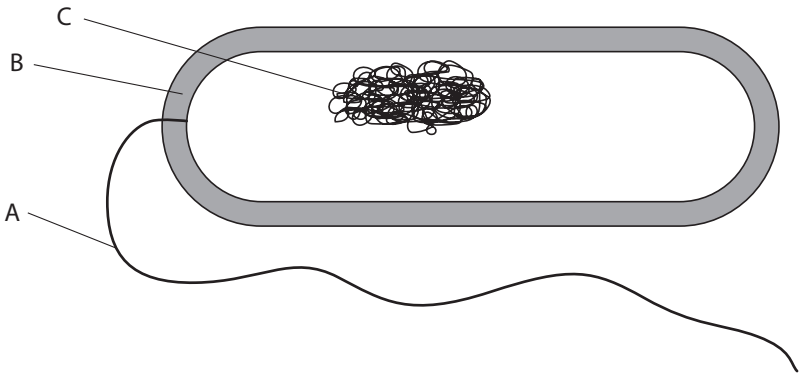
Further practice · Attempt part a.ii only.

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8 The diagram shows a bacterial cell.



(a) (i) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

(1)

.....

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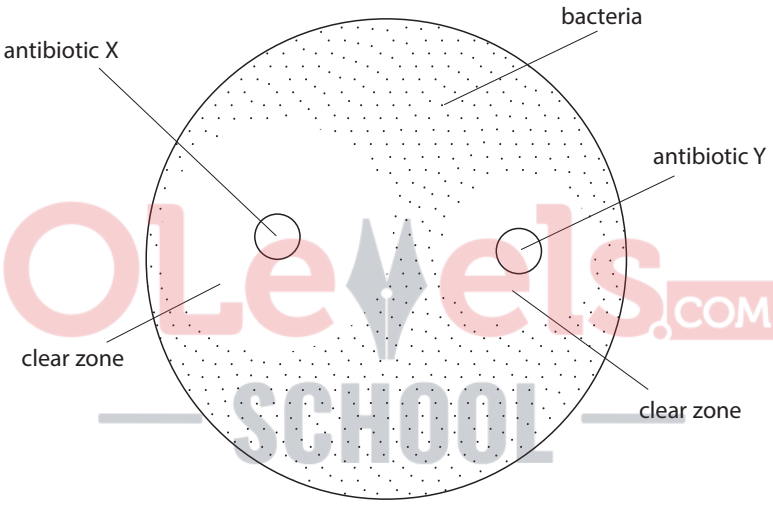
5. Question 8 · part a.ii · 1 mark

June 2023 Paper 1R

Further practice · Attempt part a.ii only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days



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(i) Describe what is meant by the term **antibiotic**.

(2)

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5. Question 8 · part a.ii · 1 mark · Working and answer

Working and answer

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6. Question 3 · part a · 1 mark

November 2023 Paper 2

Further practice · Attempt part a only.

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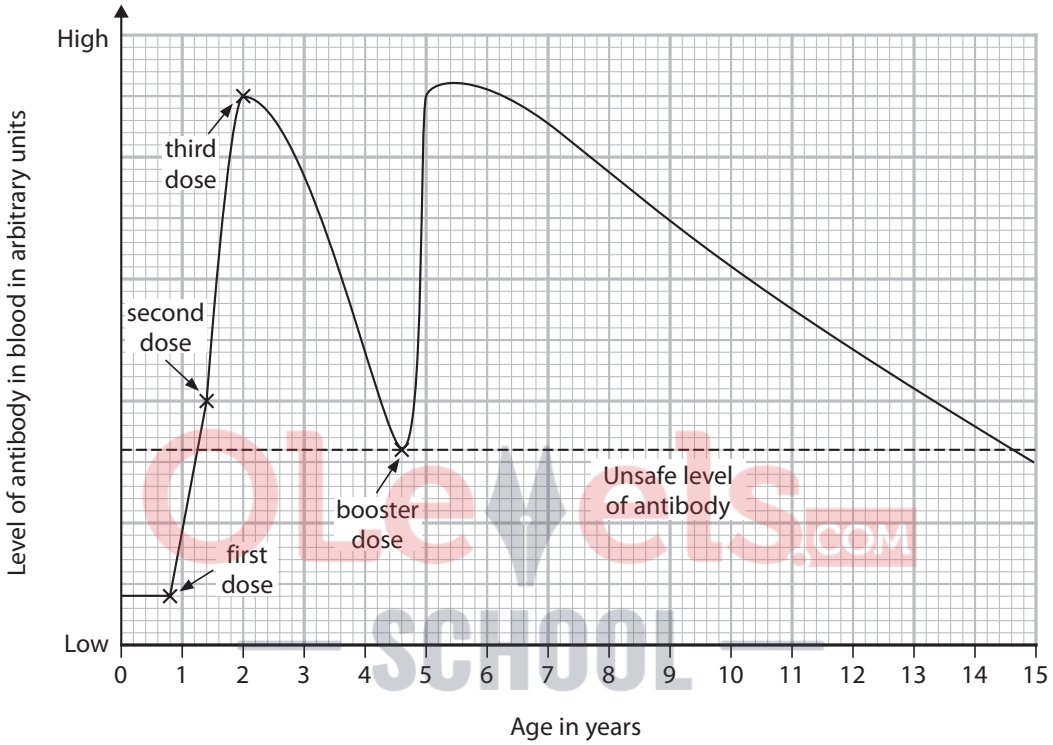
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3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

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6. Question 3 · part a · 1 mark · Working and answer

Working and answer

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7. Question 4 · part a · 1 mark

June 2025 Paper 1

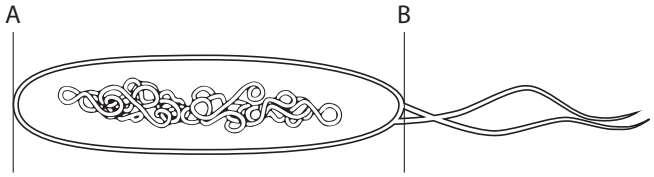
Further practice · Attempt part a only.

4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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7. Question 4 · part a · 1 mark · Working and answer

Working and answer

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8. Question 3 · part a · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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8. Question 3 · part a · 1 mark · Working and answer

Working and answer

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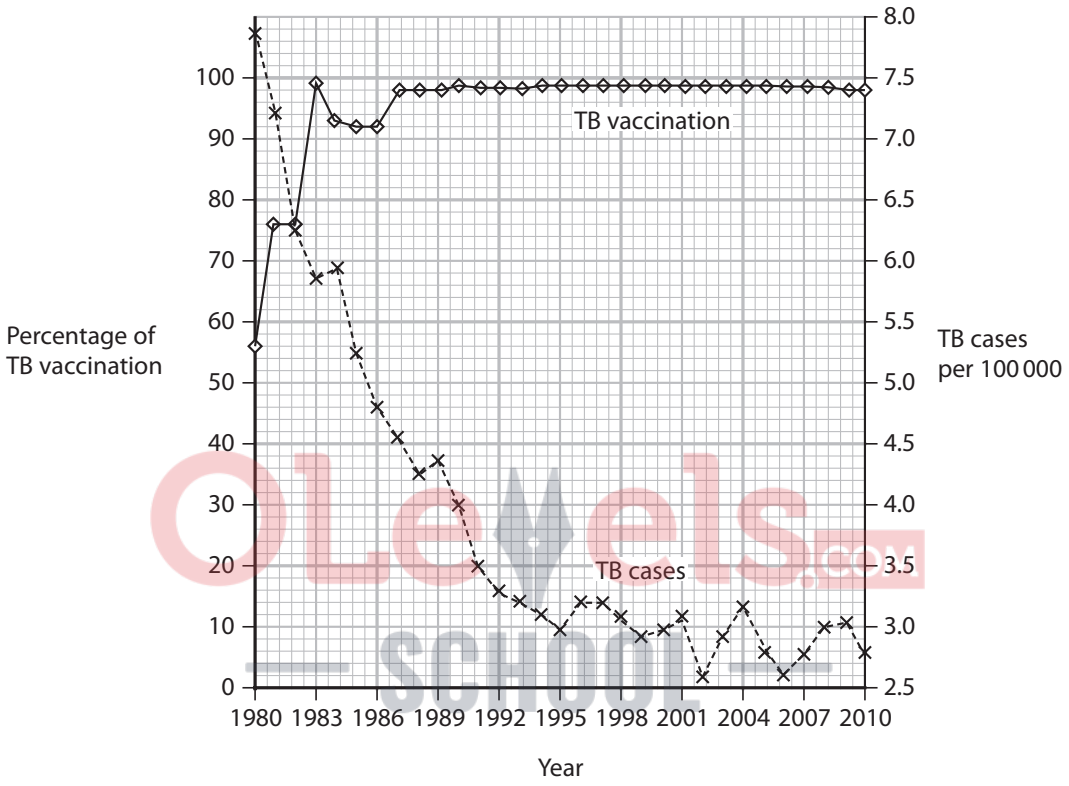
9. Question 4 · part a.i · 1 mark

June 2021 Paper 1

Further practice · Attempt part a.i only.

4 The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.



(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

(1)

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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9. Question 4 · part a.i · 1 mark

June 2021 Paper 1

Further practice · Attempt part a.i only.

(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

(4)

(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

(3)

number of cases = ...

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9. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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10. Question 7 · part a · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a only.

7 Cholera is a disease that causes chronic water loss from the body.

(a) Name the type of microorganism that causes cholera.

(1)

(b) Describe how cholera is transmitted from one person to another.

(2)

(c) Explain how a person with cholera can be treated for the disease.

(4)

(Total for Question 7 = 7 marks)

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10. Question 7 · part a · 1 mark · Working and answer

Working and answer

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11. Question 8 · part a.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.i only.

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8 (a) Cholera is a disease that kills approximately 95 000 people each year worldwide.

(i) Name the type of pathogen that causes cholera.

(1)

(ii) State how cholera is transmitted from one person to another.

(1)

(b) Describe a method used to treat a person with cholera.

(3)

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(Total for Question 8 = 5 marks)

Working and answer

12. Question 1 · part a · 3 marks

June 2024 Paper 2

Further practice · Attempt part a only.

1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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12. Question 1 · part a · 3 marks · Working and answer

Working and answer

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13. Question 8 · part b.i · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.i only.

8 (a) (i) Describe the structure of a virus.

(2)

(ii) Viruses attach to cells of various body tissues.

The virus uses the host cell's genetic material to produce new viral proteins.

Describe how the host cell carries out protein synthesis to produce new viral proteins.

(5)

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13. Question 8 · part b.i · 1 mark

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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(b) In 2010, AIDS-related illnesses caused the death of 1.4 million people.

In 2021, AIDS-related illnesses caused the death of 650 000 people.

(i) Name the microorganism that causes AIDS.

(1)

.....

(ii) Calculate the percentage decrease in the number of people dying from AIDS-related illnesses between 2010 and 2020.

(2)

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percentage decrease = ... %

(Total for Question 8 = 10 marks)

Working and answer

14. Question 1 · part a · 6 marks

November 2020 Paper 2

Further practice · Attempt part a only.

1 (a) Different types of organisms cause diseases.

Diseases are transmitted in different ways.

The table lists three diseases.

Complete the table by giving the type of organism that causes each disease and the method of transmission for each disease.

(6)

Name of disease	Type of organism	Method of transmission
malaria		
poliomyelitis		
typhoid		

(b) Explain why it is not possible to treat athlete's foot using antibiotics.

(2)

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(Total for Question 1 = 8 marks)

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14. Question 1 · part a · 6 marks · Working and answer

Working and answer

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Practice workbook · Identify bacterial cell structures and compare with human cells

Labelling bacterial cells and comparing their parts and functions with human cells such as skin cells or sperm.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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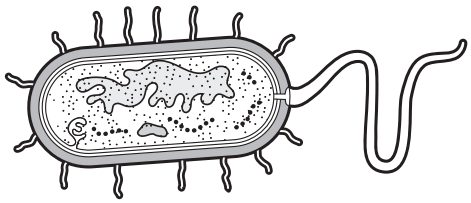
Type ID: HBI-BACTERIAL-CELL-STRUCTURE

1. Question 1 · part a.ii · 2 marks

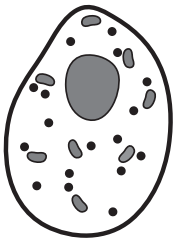
June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

1 (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

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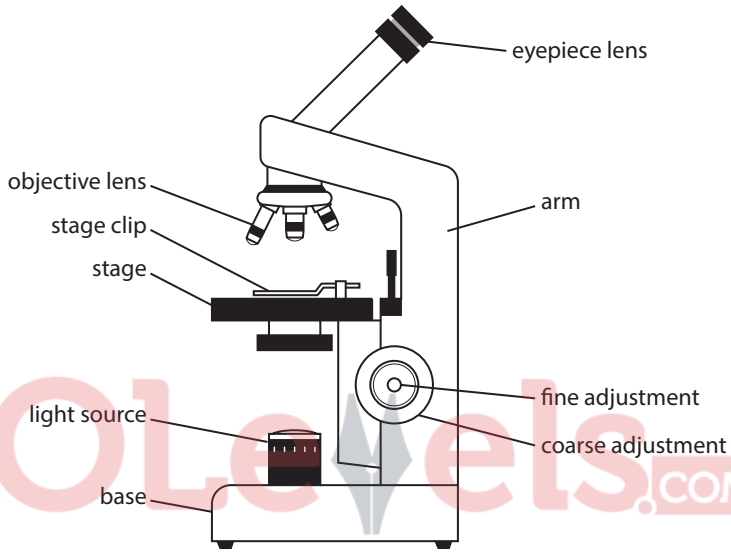
1. Question 1 · part a.ii · 2 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.ii only.

(b) Many structures in cells can be viewed more clearly using a light microscope.

The diagram shows a light microscope.



(i) Draw **one** straight line from each part of the light microscope to its function.

(3)

Part of microscope	Function
objective lens	holds the microscope slide in place
stage clip	moves the stage to produce a clear, focused image
fine adjustment	magnifies the sample
	keeps the microscope steady

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1. Question 1 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

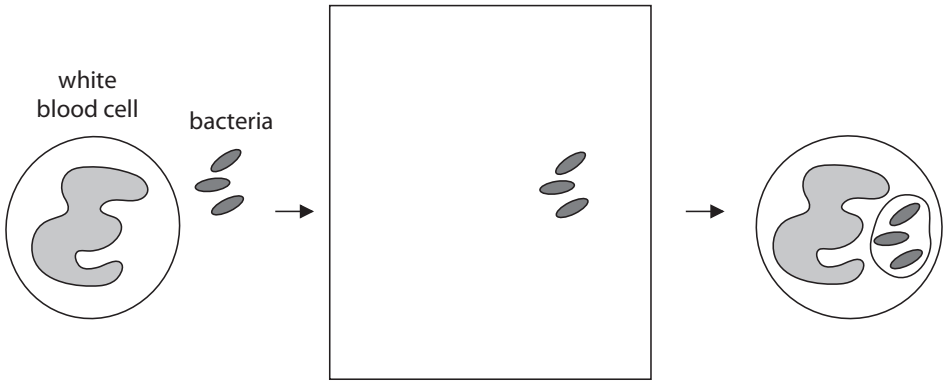
2. Question 2 · part b · 3 marks

June 2019 Paper 1

Further practice · Attempt part b only.

2 (a) Bacteria can cause disease in humans.

The diagram shows how one type of white blood cell helps to defend the body against disease.



(i) Complete the diagram by drawing the shape of the white blood cell at stage 2.

(1)

(ii) The box lists words associated with bacteria and disease.

acids	enzymes	erythrocytes	lymphocytes	phagocytes	toxins

Use words from the box to complete the sentences.

(2)

White blood cells called ..... engulf bacteria.

These white blood cells contain ..... to digest bacteria.

(iii) One way that white blood cells defend the body from disease is shown in the diagram.

State another way in which white blood cells defend the body against disease.

(1)

.....

.....

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2. Question 2 · part b · 3 marks

June 2019 Paper 1

Further practice · Attempt part b only.

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(b) The table lists structures found in some cells.

Place ticks in boxes to show which structures are found in bacterial cells and which are found in human skin cells.

One row has been completed for you.

(3)

Structures	Bacterial cell	Human skin cell
nucleus		
DNA		
cytoplasm	✓	✓
cell wall		

(c) Viruses can also cause diseases in humans.

Many viruses contain RNA as their genetic material.

Which statement describes the structure of RNA?

(1)

☒ **A** a double-stranded helix containing the bases ATGC

☒ **B** a double-stranded helix containing the bases AUGC

☒ **C** a single-stranded helix containing the bases ATGC

☒ **D** a single-stranded helix containing the bases AUGC

(Total for Question 2 = 8 marks)

2. Question 2 · part b · 3 marks · Working and answer

Working and answer

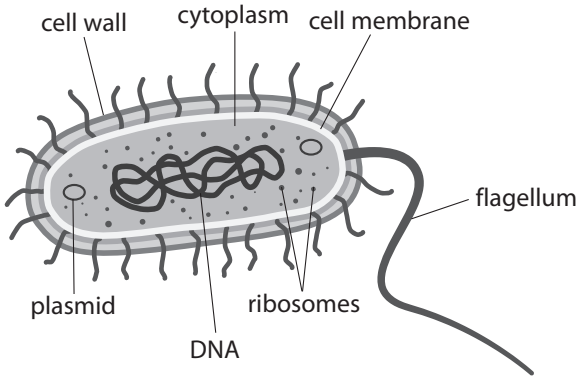
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3. Question 3 · part a.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

3 (a) The diagram shows a type of bacterial cell.



(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

(3)

Structure	Function
cell membrane	protects against pathogens
	helps the cell to move
ribosomes	makes proteins
	holds the genetic code
flagellum	
	controls which substances enter and exit the cell

(ii) State two structural differences between the bacterial cell and a sperm cell.

(2)

1 .....

.....

2 .....

.....

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3. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

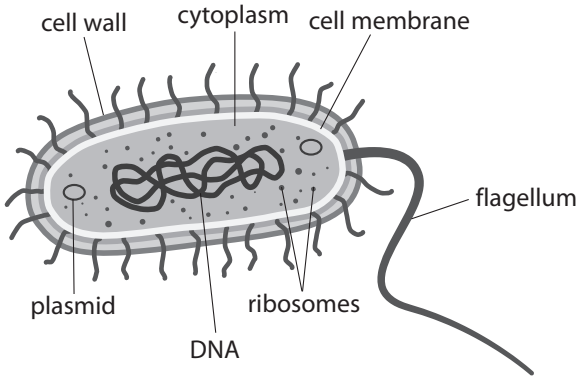
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4. Question 3 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

3 (a) The diagram shows a type of bacterial cell.



(i) The boxes give some structures found in the bacterial cell and some functions of cell structures.

Draw one straight line from each structure to its correct function.

(3)

Structure	Function
cell membrane	protects against pathogens
	helps the cell to move
ribosomes	makes proteins
	holds the genetic code
flagellum	controls which substances enter and exit the cell

(ii) State two structural differences between the bacterial cell and a sperm cell.

(2)

1 .....

.....

2 .....

.....

8

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4. Question 3 · part a.ii · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.ii only.

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(b) Many bacteria absorb oxygen and nutrients from their environment. Oxygen and nutrients are used by bacteria in aerobic respiration.

(i) Which process transports oxygen into bacterial cells?

(1)

- ☒ **A** active transport
- ☒ **B** diffusion
- ☒ **C** excretion
- ☒ **D** osmosis

(ii) Complete the word equation for aerobic respiration.

(2)

oxygen + ... → ..... + water

(iii) Explain why respiration is important for bacteria.

(2)

.....

.....

.....

.....

(Total for Question 3 = 10 marks)

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4. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 4 · part a.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part a.i only.

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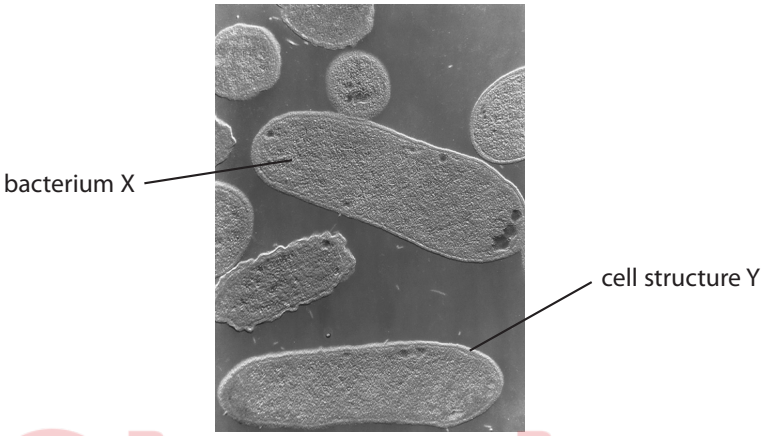
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4 Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.



(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

(1)

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

(2)

actual length = ... µm

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5. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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6. Question 8 · part a.i · 3 marks

June 2023 Paper 1R

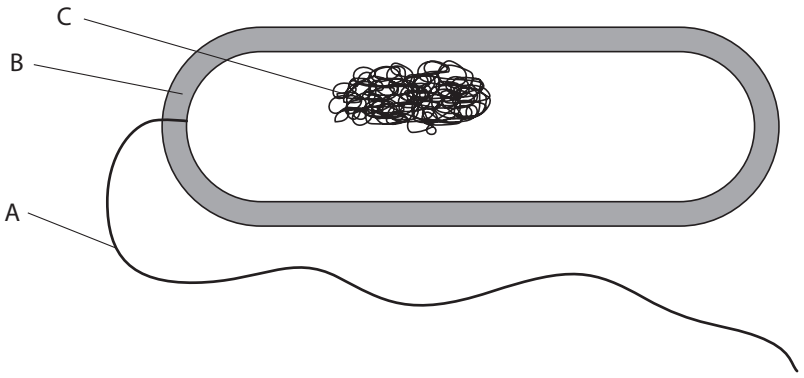
Further practice · Attempt part a.i only.

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8 The diagram shows a bacterial cell.



(a) (i) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

(1)

.....

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6. Question 8 · part a.i · 3 marks · Working and answer

Working and answer

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7. Question 1 · part a.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

1 (a) The table shows some cell structures and their functions.

(i) Place one tick (✓) in each row to show the function of each cell structure.

(3)

Cell structure	Function		
	makes proteins	**contains DNA**	**allows substances in and out of cells**
nucleus			
ribosomes			
cell membrane			

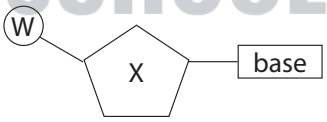
(ii) Give a structure found in bacterial cells.

(1)

.....

(b) (i) DNA is made of two chains of nucleotides.

The diagram shows one nucleotide.



Name part W and part X.

(2)

W .....

X .....

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7. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

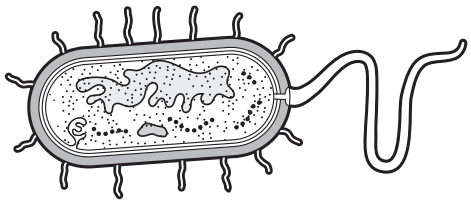
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8. Question 1 · part a.i · 2 marks

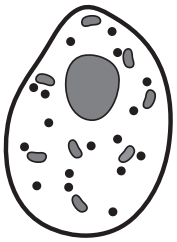
June 2024 Paper 1R

Further practice · Attempt part a.i only.

1 (a) The diagrams show a bacterial cell and a human skin cell.



Bacterial cell



Human skin cell

(i) Give **two** similarities between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

(ii) Give **two** differences between the bacterial cell and the human skin cell.

(2)

1 .....

.....

2 .....

.....

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2

8. Question 1 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 4 · part b.i · 3 marks

June 2025 Paper 1

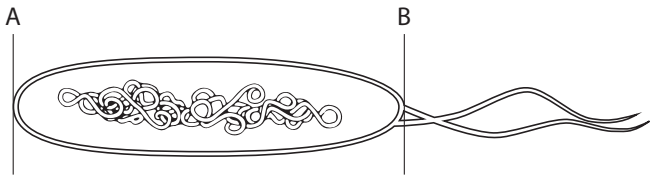
Further practice · Attempt part b.i only.

4 (a) State what is meant by the term **pathogen**.

(1)

.....

(b) The diagram shows a pathogenic bacterium.



(i) Label **three** structures shown on the diagram.

(3)

(ii) The actual length of the bacterium between points A and B is 2.0 µm.

Calculate the magnification of the diagram.

[1 mm = 1000 µm]

(3)

magnification = × .....

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9. Question 4 · part b.i · 3 marks · Working and answer

Working and answer

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Practice workbook · Describe how a named infection spreads, shows itself and is prevented

Transmission routes, reservoirs, incubation, symptoms and detection, and prevention measures for Ebola, HIV, cholera, gonorrhoea, pinworm and diarrhoea, including blood-borne risks.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DISEASE-TRANSMISSION-PREVENTION

1. Question 3 · part b · 3 marks

June 2019 Paper 1R

Worked-example retry · Attempt part b only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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1. Question 3 · part b · 3 marks · Working and answer

Working and answer

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2. Question 7 · part b · 2 marks

June 2022 Paper 1R

Further practice · Attempt part b only.

7 Cholera is a disease that causes chronic water loss from the body.

(a) Name the type of microorganism that causes cholera.

(1)

(b) Describe how cholera is transmitted from one person to another.

(2)

(c) Explain how a person with cholera can be treated for the disease.

(4)

(Total for Question 7 = 7 marks)

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2. Question 7 · part b · 2 marks · Working and answer

Working and answer

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3. Question 8 · part a.ii · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

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8 (a) Cholera is a disease that kills approximately 95 000 people each year worldwide.

(i) Name the type of pathogen that causes cholera.

(1)

(ii) State how cholera is transmitted from one person to another.

(1)

(b) Describe a method used to treat a person with cholera.

(3)

(Total for Question 8 = 5 marks)

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Working and answer

4. Question 1 · part b · 4 marks

June 2024 Paper 2

Further practice · Attempt part b only.

DO NOT WRITE IN THIS

1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

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(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

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The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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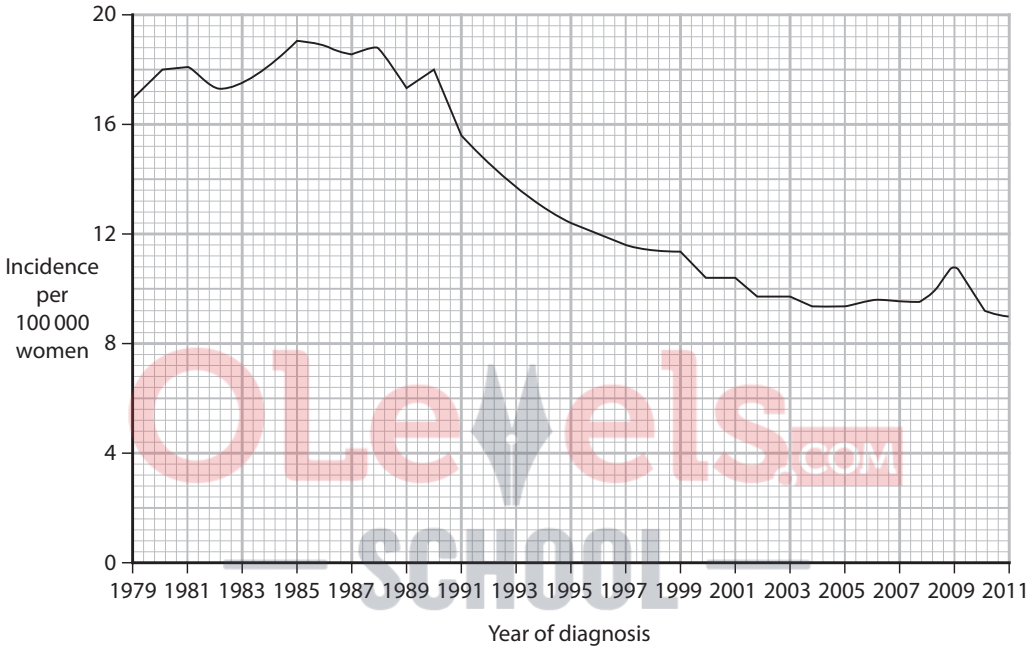
4. Question 1 · part b · 4 marks

June 2024 Paper 2

Further practice · Attempt part b only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.



(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

(2)

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P 7 4 6 3 0 A 0 4 2 4

4. Question 1 · part b · 4 marks · Working and answer

Working and answer

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5. Question 4 · part a · 2 marks

November 2024 Paper 2

Further practice · Attempt part a only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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5. Question 4 · part a · 2 marks

November 2024 Paper 2

Further practice · Attempt part a only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

5. Question 4 · part a · 2 marks · Working and answer

Working and answer

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6. Question 4 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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6. Question 4 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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6. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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7. Question 4 · part e · 2 marks

January 2020 Paper 2

Further practice · Attempt part e only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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7. Question 4 · part e · 2 marks

January 2020 Paper 2

Further practice · Attempt part e only.

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(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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7. Question 4 · part e · 2 marks

January 2020 Paper 2

Further practice · Attempt part e only.

(e) Suggest how an infected person and other people in the house can help prevent the spread of a pinworm infection (lines 13 to 14).

(2)

.....

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(Total for Question 4 = 12 marks)

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Working and answer

8. Question 2 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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8. Question 2 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

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8. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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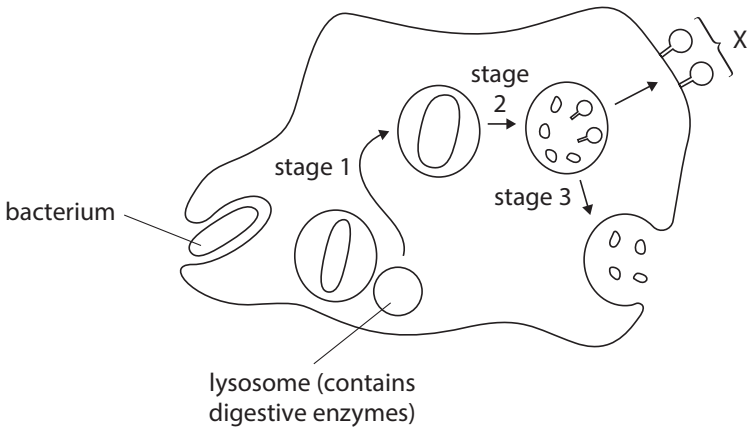
9. Question 6 · part b.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

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6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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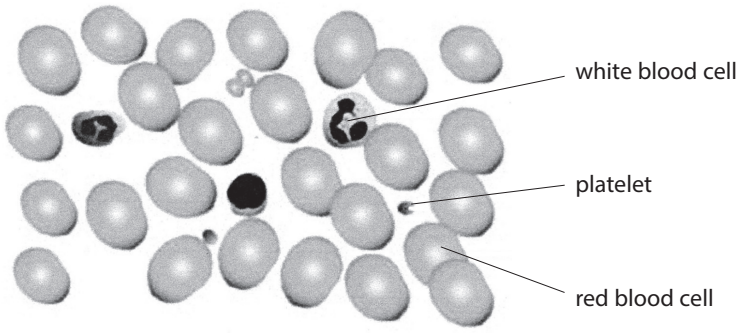
9. Question 6 · part b.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.



(i) Describe how the slide of human blood could be prepared.

(3)

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9. Question 6 · part b.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Discuss the accuracy of the student's diagram.

(4)

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(iii) Explain why the student would be advised to use a prepared slide rather than making one from their own blood.

(2)

(Total for Question 6 = 15 marks)

9. Question 6 · part b.iii · 2 marks · Working and answer

Working and answer

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10. Question 3 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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10. Question 3 · part e · 2 marks

June 2019 Paper 1R

Further practice · Attempt part e only.

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(d) Oral rehydration is a method used to treat the symptoms of Ebola.

Explain why this method is an effective treatment.

(3)

(e) Fruit bats are thought to be a natural reservoir for the Ebola pathogen.

Suggest what is meant by the term **reservoir**.

(2)

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(Total for Question 3 = 12 marks)

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10. Question 3 · part e · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 3 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

3 Ebola is a disease that has become an epidemic in some countries.

(a) Which of these causes Ebola?

(1)

- ☒ **A** a bacterium
- ☒ **B** a mosquito
- ☒ **C** a protozoan
- ☒ **D** a virus

(b) State three precautions that can be taken to prevent the spread of Ebola.

(3)

1 .....

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2 .....

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3 .....

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(c) Symptoms of Ebola only appear two to three weeks after infection.

Explain why the symptoms do not appear immediately.

(3)

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11. Question 3 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

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(d) Oral rehydration is a method used to treat the symptoms of Ebola.

Explain why this method is an effective treatment.

(3)

(e) Fruit bats are thought to be a natural reservoir for the Ebola pathogen.

Suggest what is meant by the term **reservoir**.

(2)

(Total for Question 3 = 12 marks)

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11. Question 3 · part c · 3 marks · Working and answer

Working and answer

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12. Question 4 · part a.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part a.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

Pinworm infections are highly contagious and occur most commonly in children under the age of five years. Infections occur when eggs are ingested or inhaled. These eggs have been deposited onto a surface or object by an infected person. The eggs, which are very tiny, can survive on these surfaces for several weeks.

One method of detecting a pinworm infection is to carry out a tape test. This test consists of sticking a piece of transparent tape to the anus in the morning. If an infection is detected then medication can be taken to treat it. In addition to taking medication, it is important that the infected person and other people in the house follow strict procedures to eliminate pinworm eggs and to prevent the spread of the infection.

In rare cases where infections are not treated, large populations of pinworms can grow in the intestines. This can lead to nutritional deficiencies in the infected person, which cause further symptoms that have a more serious effect on health.

(a) A pinworm infection causes itching.

(i) Give one other symptom of a pinworm infection.

(1)

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(ii) Explain why itching plays an important role in the spread of a pinworm infection (lines 4 to 5).

(2)

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12. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 4 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Pinworms are small parasites that live in the small intestine and colon of humans. They emerge at night and lay their eggs in the anal region. In addition to laying over 10 000 eggs at one time, pinworms also secrete a substance that causes a very strong itching sensation around the anus. Although this is particularly uncomfortable for the host, itching plays an important role in the spread of infection.

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13. Question 4 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

(b) Suggest how the small intestine provides perfect conditions for the growth of pinworm populations.

(2)

(c) Pinworms can cause nutritional deficiencies (line 16).

(i) State what is meant by the term **nutritional deficiency**.

(1)

(ii) State one nutritional deficiency that could be caused by a pinworm infection.

(1)

(iii) Suggest why a large population of pinworms in a person's intestines can lead to nutritional deficiency.

(2)

(d) Suggest how the transparent tape test can be used to detect a pinworm infection (lines 10 to 11).

(1)

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13. Question 4 · part d · 1 mark

January 2020 Paper 2

Further practice · Attempt part d only.

(e) Suggest how an infected person and other people in the house can help prevent the spread of a pinworm infection (lines 13 to 14).

(2)

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(Total for Question 4 = 12 marks)

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Working and answer

Practice workbook · Explain antigen-antibody specificity and how monoclonal antibodies are made

What antigens are, why each antibody binds one antigen, reactions to self or foreign proteins, and producing monoclonal antibodies from fused cells.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ANTIGEN-ANTIBODY-SPECIFICITY

1. Question 4 · part b · 3 marks

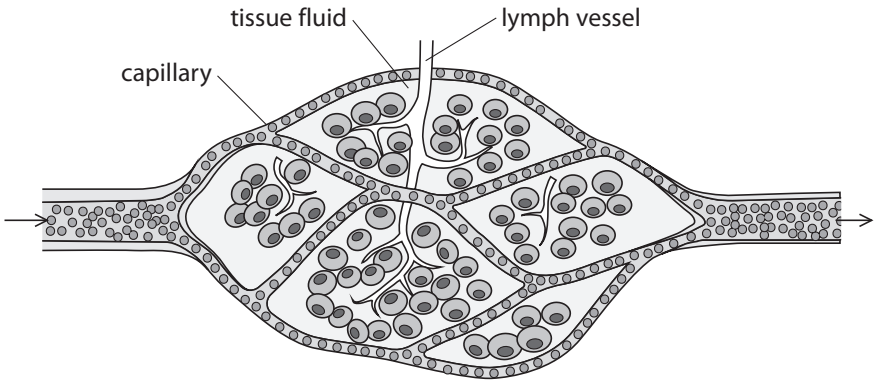
January 2023 Paper 2

Worked-example retry · Attempt part b only.

4 (a) When blood flows into a network of capillaries, tissue fluid is produced.

The tissue fluid supplies the surrounding cells with molecules.

The diagram shows a network of capillaries and surrounding cells.



(i) Give two differences between the composition of tissue fluid and the composition of blood entering the capillary network.

(2)

1 .....

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2 .....

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(ii) Describe how tissue fluid is formed.

(4)

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1. Question 4 · part b · 3 marks

January 2023 Paper 2

Worked-example retry · Attempt part b only.

(iii) Lymphoedema occurs when tissue fluid builds up causing swelling.

This is more common in the legs and arms.

One treatment for this is to increase physical activity, resulting in more muscle contraction.

Explain why increased muscle contraction could reduce the swelling caused by the build up of tissue fluid.

(2)

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(b) Lymphocytes, found in lymph nodes, produce antibodies in response to specific antigens.

Explain why the reaction between an antigen and an antibody is specific.

(3)

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(Total for Question 4 = 11 marks)

1. Question 4 · part b · 3 marks · Working and answer

Working and answer

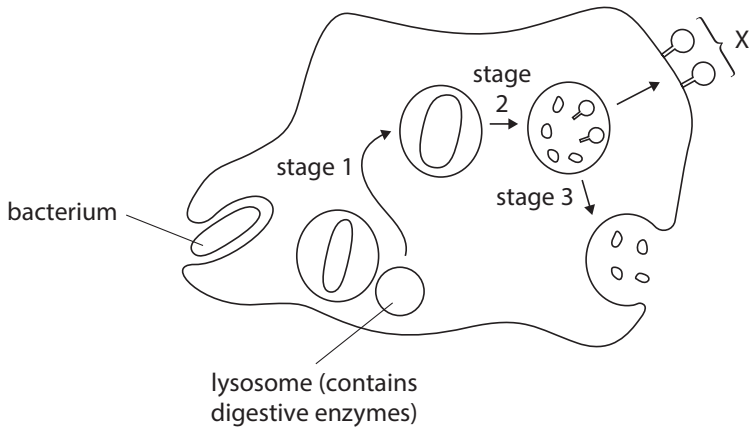
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2. Question 6 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

6 The diagram shows a white blood cell engulfing a bacterium and the stages that follow.



(a) (i) State the name of the process used by white blood cells to engulf bacteria.

(1)

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

(3)

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

(2)

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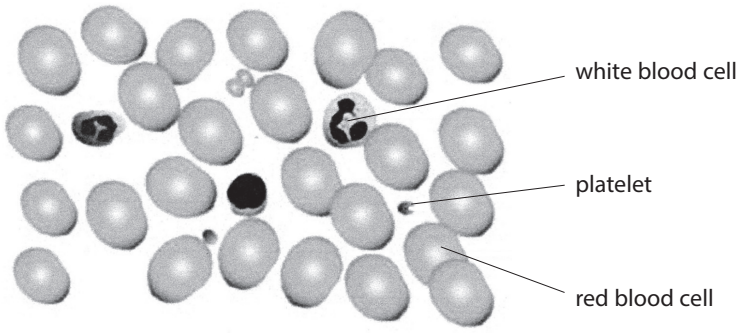
2. Question 6 · part a.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part a.iii only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.



(i) Describe how the slide of human blood could be prepared.

(3)

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2. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

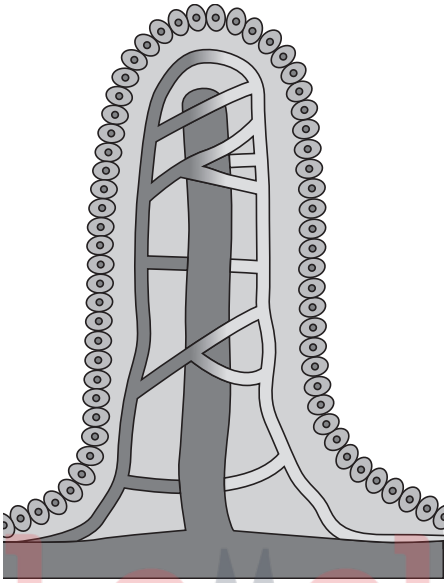
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3. Question 7 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

7 The diagram shows a villus in the small intestine of the digestive system.



(a) Explain how the villus is adapted to its function in the digestive system.

(3)

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3. Question 7 · part b.i · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.i only.

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(b) (i) Coeliac disease is an autoimmune disease.

The immune system of people with coeliac disease attacks the body's own tissue and damages the lining of the small intestine.

Describe how the immune system of people with coeliac disease damages the lining of the small intestine.

(2)

(ii) Explain how coeliac disease affects the absorption of nutrients in the small intestine.

(3)

(Total for Question 7 = 8 marks)

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3. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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4. Question 7 · part d · 2 marks

June 2025 Paper 2R

Further practice · Attempt part d only.

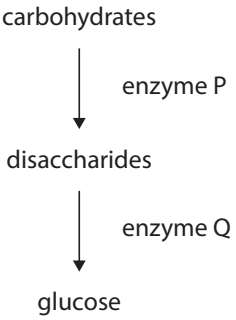
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7 (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.



Name enzyme P.

(1)

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(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

(2)

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.....

(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

(1)

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4. Question 7 · part d · 2 marks

June 2025 Paper 2R

Further practice · Attempt part d only.

(d) Insulin is a hormone used to treat people with diabetes.

Until recently insulin was obtained from animals for human use.

Suggest why this type of insulin can cause health complications in people with diabetes.

(2)

(e) Describe how yeast cells can be modified to produce the human form of insulin.

(2)

(Total for Question 7 = 8 marks)

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Working and answer

5. Question 8 · part a.iv · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.iv only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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5. Question 8 · part a.iv · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.iv only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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5. Question 8 · part a.iv · 1 mark · Working and answer

Working and answer

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6. Question 8 · part a.v · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.v only.

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8 (a) Read the passage below.

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(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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6. Question 8 · part a.v · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.v only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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6. Question 8 · part a.v · 2 marks · Working and answer

Working and answer

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7. Question 8 · part whole · 5 marks

January 2020 Paper 2

Further practice · Attempt part whole only.

8 Describe how monoclonal antibodies are produced.

(5)

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(Total for Question 8 = 5 marks)

7. Question 8 · part whole · 5 marks · Working and answer

Working and answer

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Practice workbook · Explain how vaccination or infection gives active immunity

How vaccines or a first infection lead to antibodies and memory cells, why boosters are given, and herd protection.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VACCINATION-ACTIVE-IMMUNITY

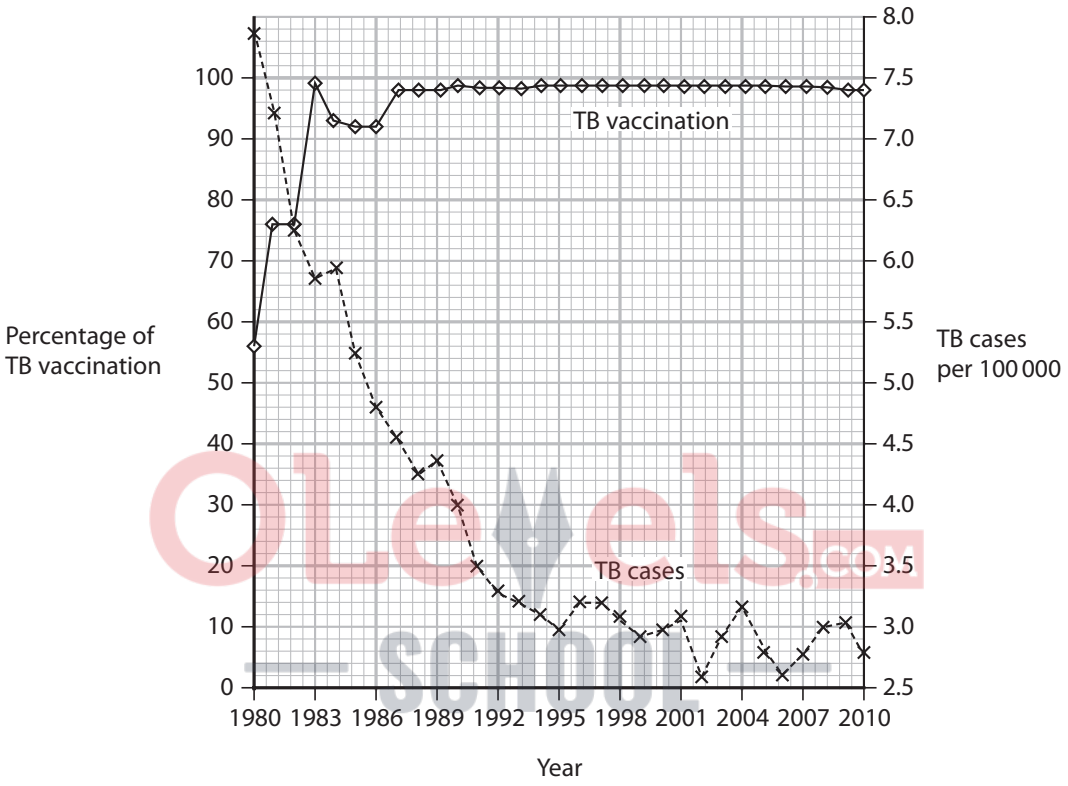
1. Question 4 · part c · 5 marks

June 2021 Paper 1

Worked-example retry · Attempt part c only.

4 The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.



(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

(1)

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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1. Question 4 · part c · 5 marks

June 2021 Paper 1

Worked-example retry · Attempt part c only.

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(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

(4)

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(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

(3)

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number of cases = ...

1. Question 4 · part c · 5 marks

June 2021 Paper 1

Worked-example retry · Attempt part c only.

(c) Explain how vaccination can give a person immunity to a disease.

(5)

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(Total for Question 4 = 13 marks)

1. Question 4 · part c · 5 marks · Working and answer

Working and answer

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2. Question 8 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

8 *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

(3)

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

Region of the world	Q	R	S	T	U
Total number of deaths from malaria	191 000	800	3800	14 400	1200
Number of deaths caused by *P. vivax*	1000	500	1400	4900	700

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

(3)

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2. Question 8 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

(1)

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

(3)

(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

(3)

(Total for Question 8 = 13 marks)

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2. Question 8 · part c · 3 marks · Working and answer

Working and answer

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3. Question 9 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

9 (a) Hepatitis is a disease of the liver.

This disease can be caused by drinking too much alcohol.

Explain the damaging effects of alcohol on the liver.

(3)

(b) Hepatitis B is one type of virus that can cause hepatitis.

Describe the reproduction of a hepatitis B virus.

(6)

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3. Question 9 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(c) Vaccinations can be used to prevent the spread of hepatitis.

Explain how vaccines can help people become immune to hepatitis.

(3)

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(Total for Question 9 = 12 marks)

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Working and answer

4. Question 3 · part b.iii · 2 marks

November 2023 Paper 2

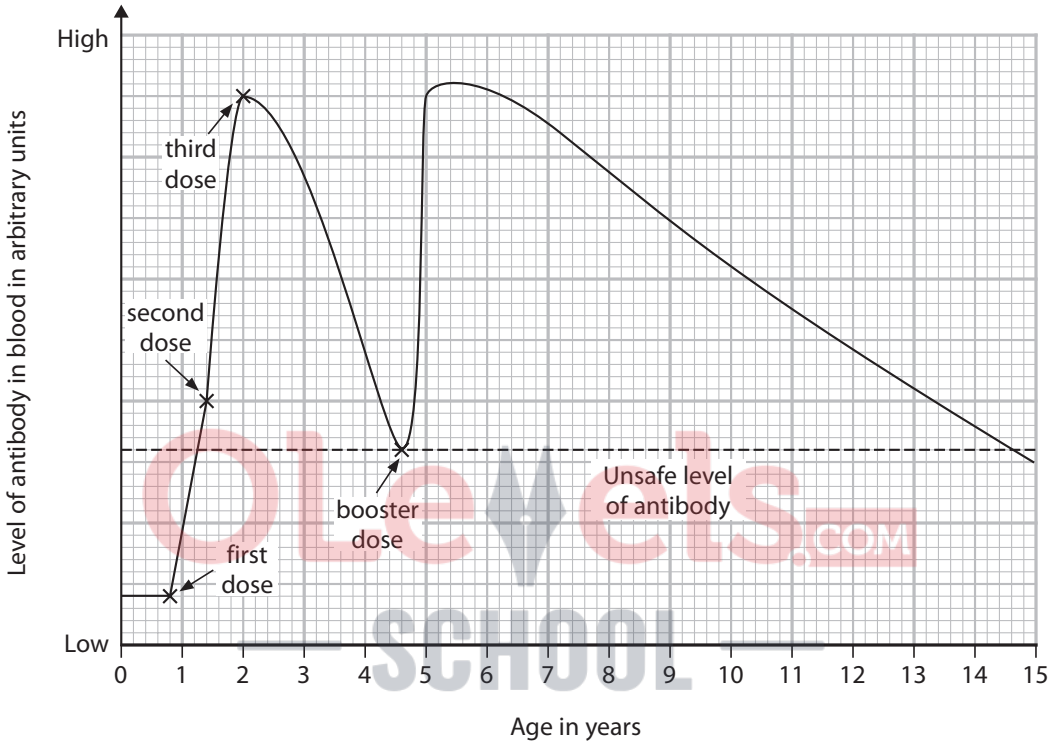
Further practice · Attempt part b.iii only.

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3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

.....

.....

4. Question 3 · part b.iii · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.iii only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

(2)

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

(2)

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(d) Explain why a booster dose is necessary.

(5)

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4. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part d · 5 marks

November 2023 Paper 2

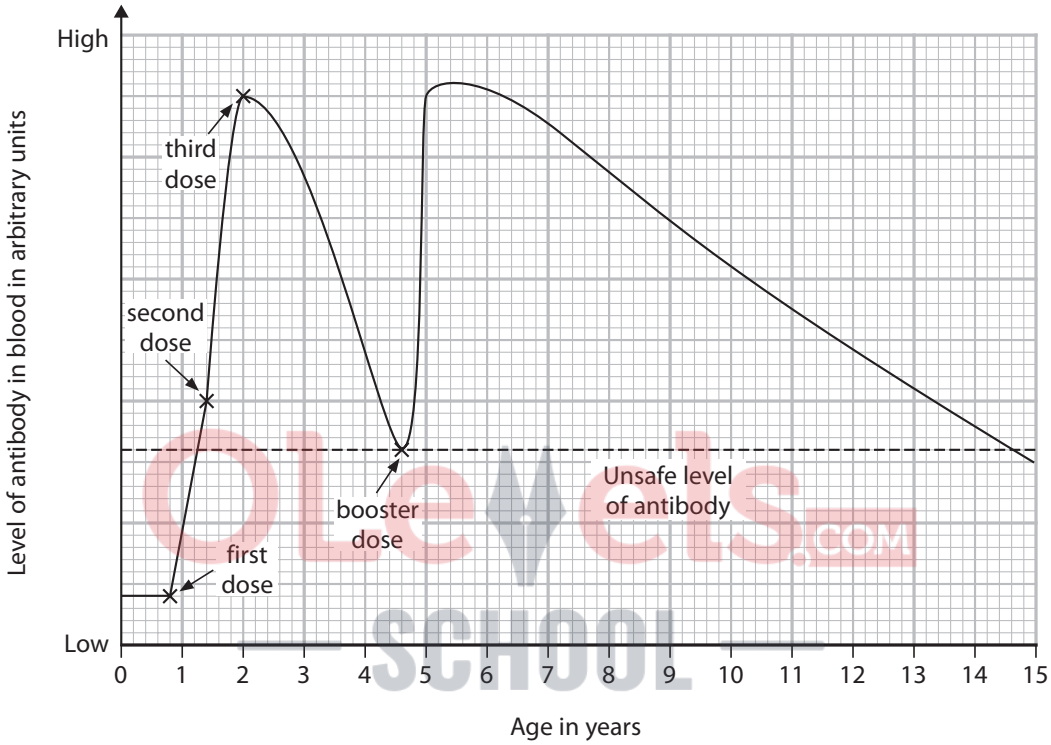
Further practice · Attempt part d only.

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(1)

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(2)

.....

.....

5. Question 3 · part d · 5 marks

November 2023 Paper 2

Further practice · Attempt part d only.

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(2)

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5. Question 3 · part d · 5 marks

November 2023 Paper 2

Further practice · Attempt part d only.

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(e) Give two differences between the effects of the first three doses and the effect of the booster dose.

(2)

1 .....

.....

2 .....

.....

(Total for Question 3 = 15 marks)

Working and answer

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6. Question 1 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

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1 (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

(3)

Disease	Type of pathogen
cholera	
AIDS	
athlete's foot	

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

(4)

hormonal	antibiotics	contaminated water	virus
sexual intercourse	painkillers	bacterium	barrier

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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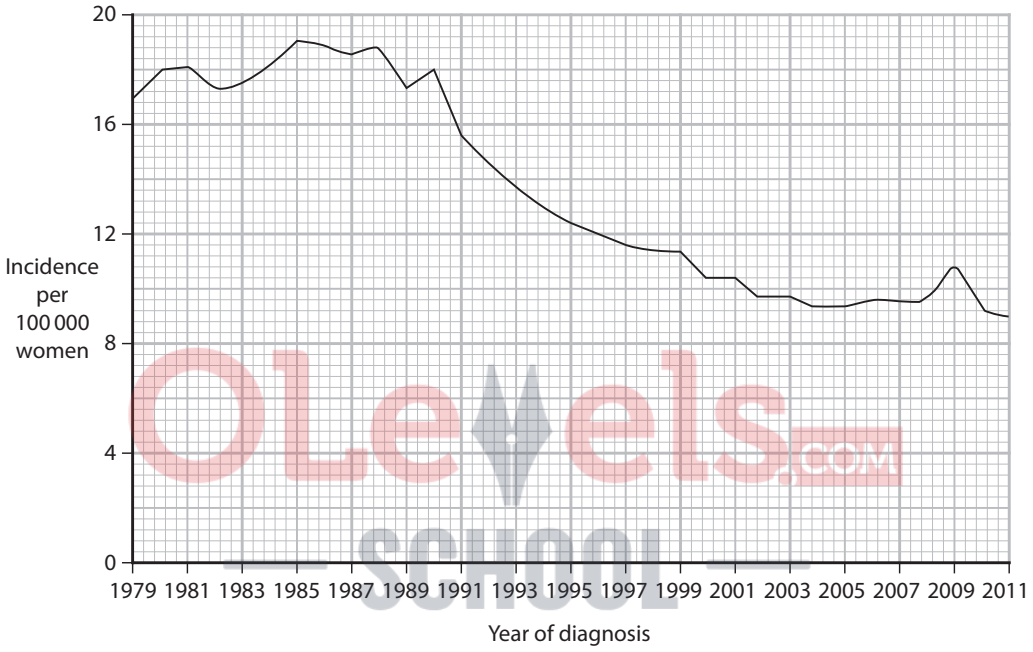
6. Question 1 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.



(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

(2)

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6. Question 1 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

(ii) Calculate the difference in the incidence per 100 000 of women with cervical cancer between 1979 and 2011.

(2)

difference = ... per 100 000

(d) Cervical cancer that is caused by a virus can now be prevented by vaccination.

The table gives information about vaccination.

Which row gives correct information about vaccination?

(1)

	Contents of vaccination	Response by the body	How the pathogen is destroyed
☒ **A**	antibodies	antigens produced	antigens destroy pathogen
☒ **B**	antibodies	white blood cells produced	white blood cells destroy pathogen
☒ **C**	antigens	antibodies produced	antibodies destroy pathogen
☒ **D**	antigens	platelets produced	platelets destroy pathogen

(Total for Question 1 = 12 marks)

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Working and answer

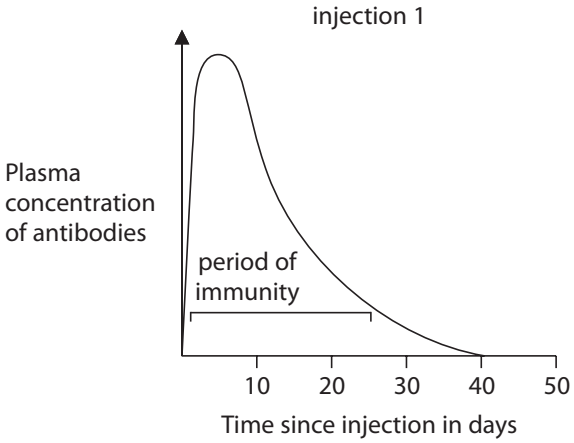
7. Question 4 · part b · 5 marks

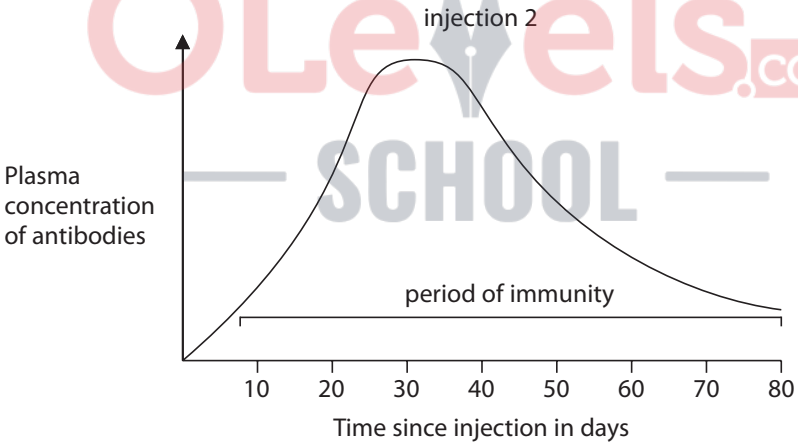
January 2022 Paper 2

Further practice · Attempt part b only.

4 (a) Artificial immunity can be produced by different types of injection.

The graphs show the plasma concentration of antibodies produced as a result of two types of injection.





(i) Identify the different forms of artificial immunity produced by these two injections.

(2)

1 .....

2 .....

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7. Question 4 · part b · 5 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(ii) Explain which injection would be better for someone who already has the disease.

(3)

(b) Describe how natural immunity to a disease can happen.

(5)

(Total for Question 4 = 10 marks)

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7. Question 4 · part b · 5 marks · Working and answer

Working and answer

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Practice workbook · Classify immunity as natural or artificial and active or passive

Distinguishing natural, artificial, active and passive immunity, including antibodies from mothers or injected donor plasma.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-CLASSIFY-IMMUNITY-TYPES

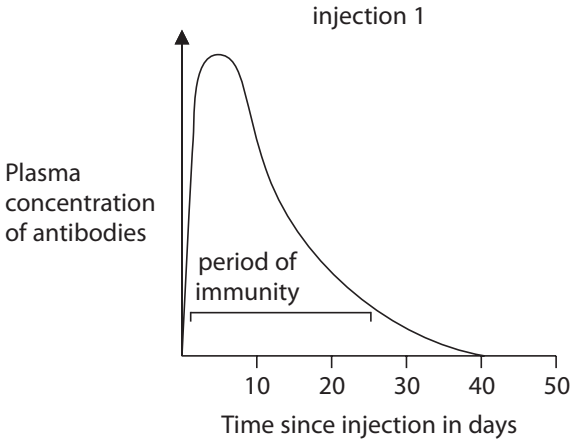
1. Question 4 · part a.i · 2 marks

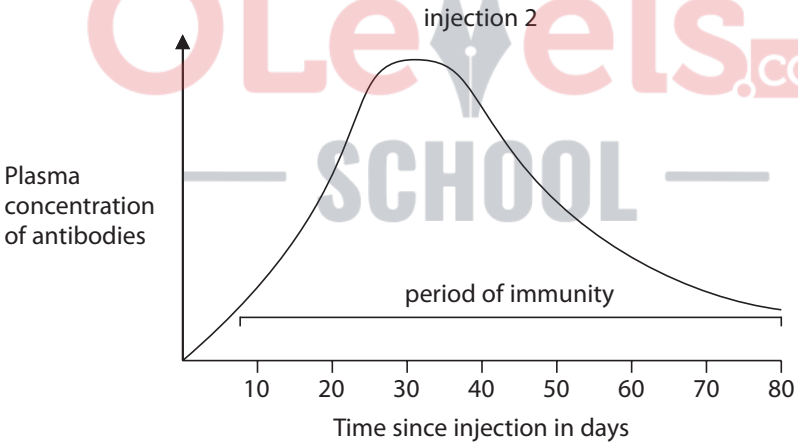
January 2022 Paper 2

Worked-example retry · Attempt part a.i only.

4 (a) Artificial immunity can be produced by different types of injection.

The graphs show the plasma concentration of antibodies produced as a result of two types of injection.





(i) Identify the different forms of artificial immunity produced by these two injections.

(2)

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2 .....

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1. Question 4 · part a.i · 2 marks

January 2022 Paper 2

Worked-example retry · Attempt part a.i only.

(ii) Explain which injection would be better for someone who already has the disease.

(3)

(b) Describe how natural immunity to a disease can happen.

(5)

(Total for Question 4 = 10 marks)

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1. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a.v · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.v only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Treatment of Ebola

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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2. Question 2 · part a.v · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.v only.

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(a) (i) Explain what is meant by an RNA virus (line 1).

(2)

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

(2)

Disease	Blood tested (✓)
anaemia	
cystic fibrosis	
gonorrhoea	
HIV	
scurvy	

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(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

(3)

2. Question 2 · part a.v · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.v only.

(iv) Suggest why donated CWB should never be frozen (line 17).

(2)

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

(2)

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

(3)

(Total for Question 2 = 14 marks)

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2. Question 2 · part a.v · 2 marks · Working and answer

Working and answer

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3. Question 7 · part b · 6 marks

November 2020 Paper 2

Further practice · Attempt part b only.

7 (a) Give two differences between whole blood and plasma.

(2)

1 .....

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2 .....

.....

(b) Ebola is an infectious disease. One symptom of Ebola is blood loss.

The World Health Organisation (WHO) has a procedure for treating Ebola.

This is the WHO's procedure.

- find a person who has recovered from Ebola
- observe this person to make sure that they are disease free for 28 days
- take blood from this person and test the blood for other diseases
- separate the plasma from the blood
- transfer this plasma to another person who has Ebola

Explain how the WHO procedure helps to treat a person who has Ebola.

(6)

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3. Question 7 · part b · 6 marks

November 2020 Paper 2

Further practice · Attempt part b only.

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(Total for Question 7 = 8 marks)

Working and answer

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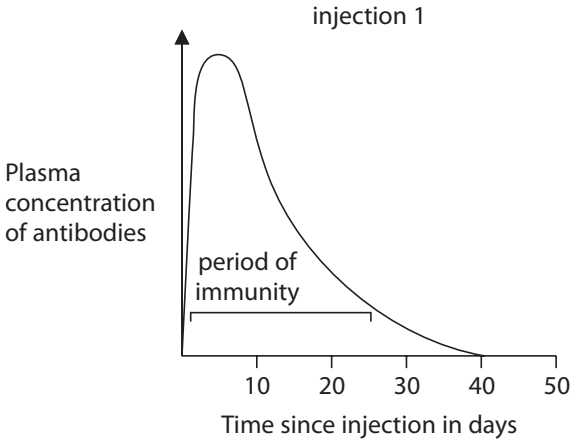
4. Question 4 · part a.ii · 3 marks

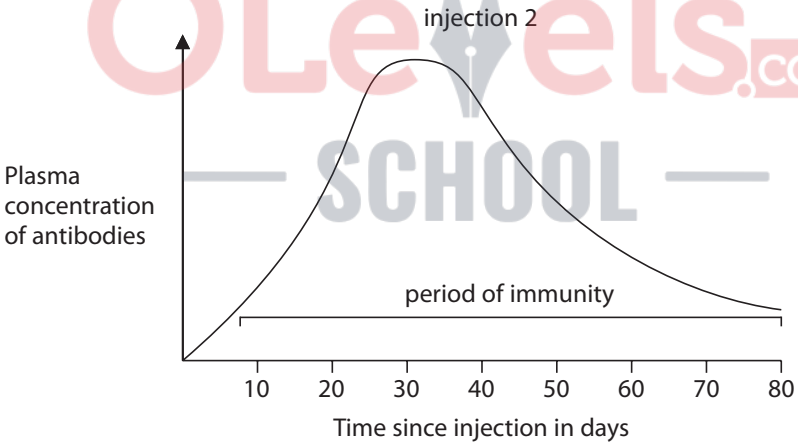
January 2022 Paper 2

Further practice · Attempt part a.ii only.

4 (a) Artificial immunity can be produced by different types of injection.

The graphs show the plasma concentration of antibodies produced as a result of two types of injection.





(i) Identify the different forms of artificial immunity produced by these two injections.

(2)

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4. Question 4 · part a.ii · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.ii only.

(ii) Explain which injection would be better for someone who already has the disease.

(3)

(b) Describe how natural immunity to a disease can happen.

(5)

(Total for Question 4 = 10 marks)

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4. Question 4 · part a.ii · 3 marks · Working and answer

Working and answer

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5. Question 3 · part b.ii · 2 marks

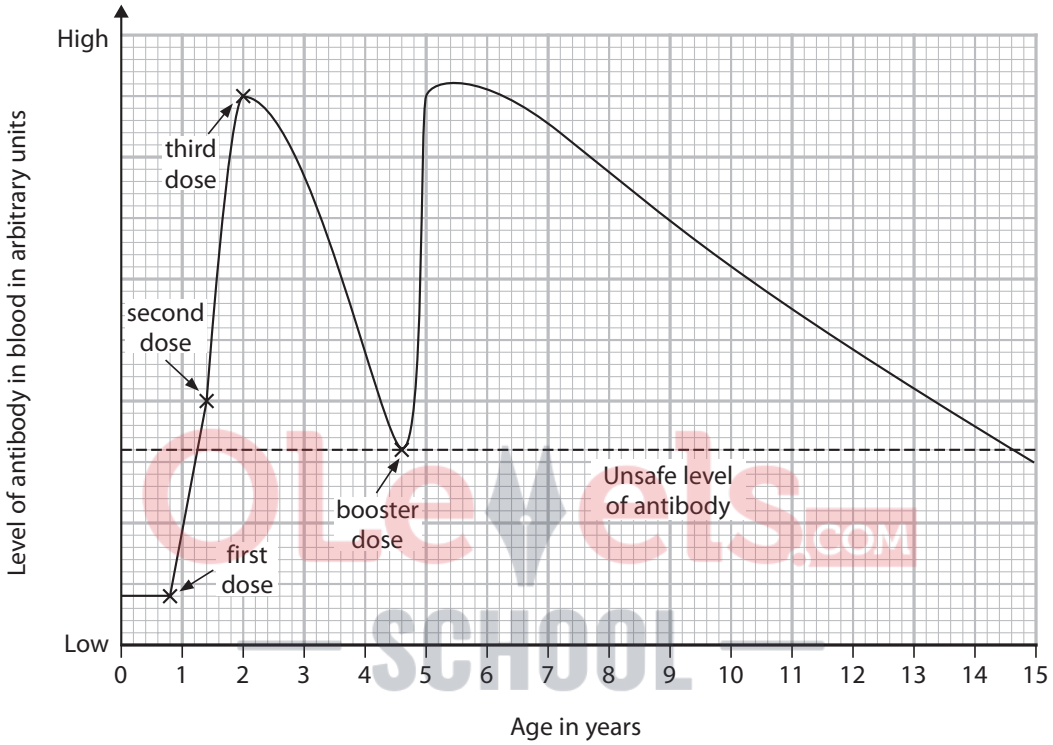
November 2023 Paper 2

Further practice · Attempt part b.ii only.

3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

.....

.....

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5. Question 3 · part b.ii · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.ii only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

(2)

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

(2)

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(d) Explain why a booster dose is necessary.

(5)

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5. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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6. Question 3 · part c · 2 marks

November 2023 Paper 2

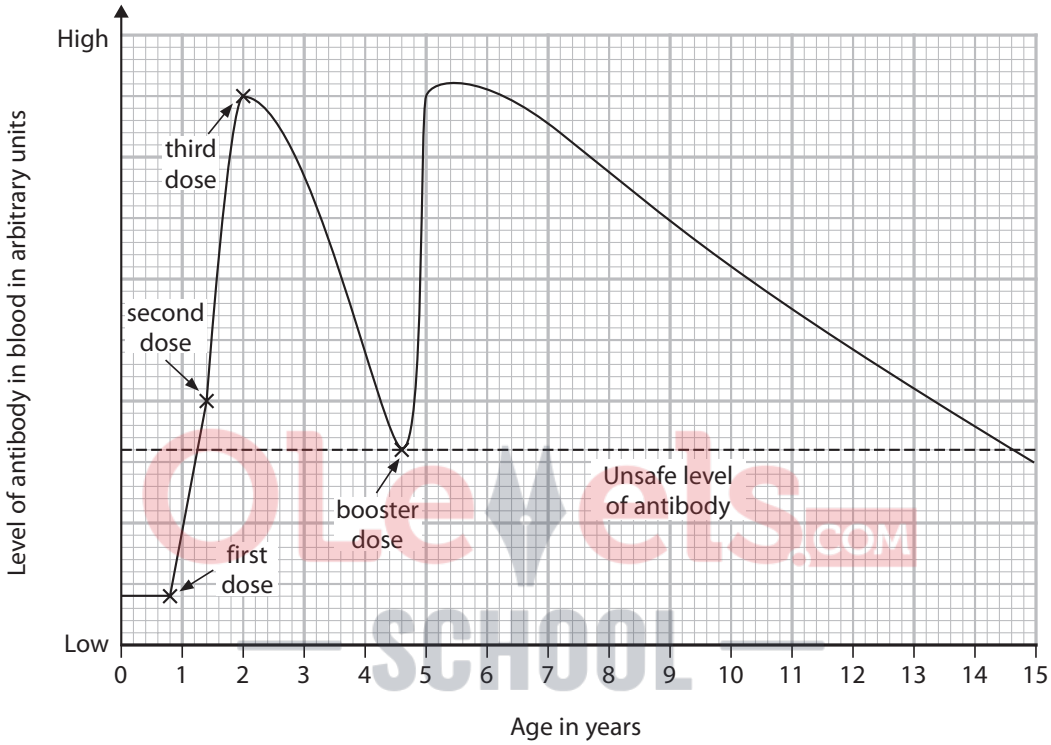
Further practice · Attempt part c only.

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3 A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.



(a) State the general name given to disease causing organisms.

(1)

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

(1)

time = ... years

(ii) State the type of immunity that vaccination produces.

(2)

.....

.....

6. Question 3 · part c · 2 marks

November 2023 Paper 2

Further practice · Attempt part c only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

(2)

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

(2)

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(d) Explain why a booster dose is necessary.

(5)

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6. Question 3 · part c · 2 marks · Working and answer

Working and answer

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7. Question 8 · part a.vi · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.vi only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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7. Question 8 · part a.vi · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.vi only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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7. Question 8 · part a.vi · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.vi only.

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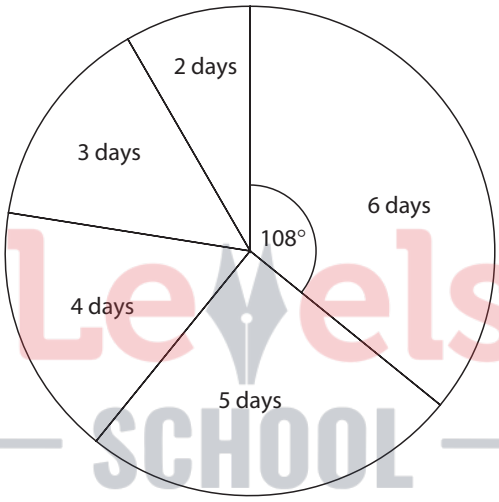
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(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.



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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

(3)

number of types of antibiotic = ...

7. Question 8 · part a.vi · 2 marks · Working and answer

Working and answer

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Practice workbook · State what antibiotics treat and explain how resistance develops

What antibiotics are and where they come from, why they fail against viruses and fungi, and selection of resistant strains such as MRSA and its consequences.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ANTIBIOTICS-AND-RESISTANCE

1. Question 1 · part b · 2 marks

November 2020 Paper 2

Worked-example retry · Attempt part b only.

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1 (a) Different types of organisms cause diseases.

Diseases are transmitted in different ways.

The table lists three diseases.

Complete the table by giving the type of organism that causes each disease and the method of transmission for each disease.

(6)

Name of disease	Type of organism	Method of transmission
malaria		
poliomyelitis		
typhoid		

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(b) Explain why it is not possible to treat athlete's foot using antibiotics.

(2)

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(Total for Question 1 = 8 marks)

1. Question 1 · part b · 2 marks · Working and answer

Working and answer

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2. Question 6 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	**MRSA**	***S. aureus***	**MRSA**
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

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2 .....

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2. Question 6 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

.....

.....

.....

.....

2. Question 6 · part b.i · 1 mark

January 2020 Paper 2

Further practice · Attempt part b.i only.

(b) (i) An infection caused by *S. aureus* is usually harmless and can be treated using antibiotics.

State one source of an antibiotic.

(1)

(ii) Describe an aseptic method that a scientist could use to find out whether an infection is caused by *S. aureus* or by MRSA.

(6)

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(Total for Question 6 = 15 marks)

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2. Question 6 · part b.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part a.ii only.

7 Various diseases are caused by microorganisms. Some diseases can be treated by using antibiotics.

(a) (i) Which of these is the name given to any microorganisms that cause disease? **(1)**

- ☒ **A** bacteria
- ☒ **B** fungi
- ☒ **C** pathogens
- ☒ **D** protozoa

(ii) Explain why it is not possible to treat Ebola using antibiotics. **(2)**

.....

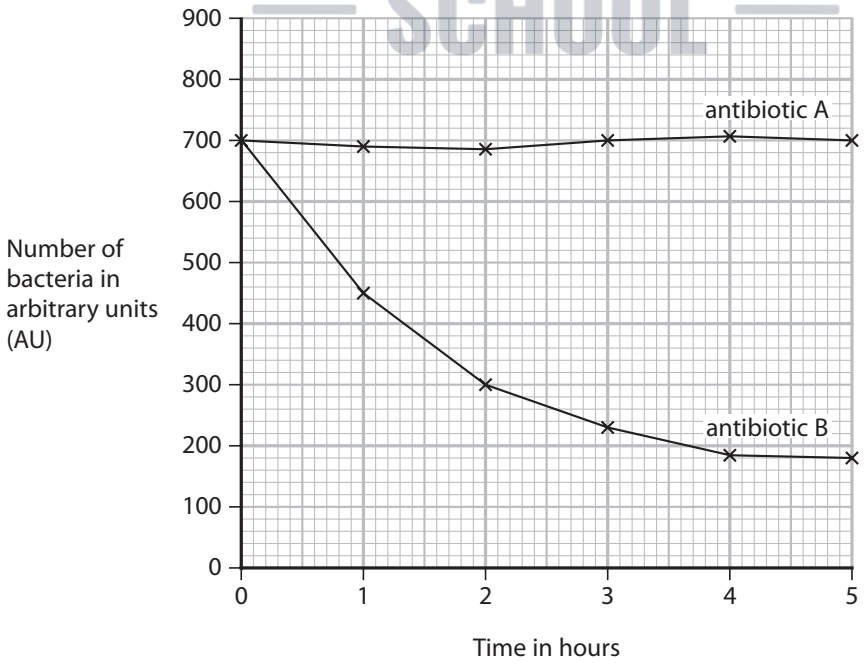
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(b) The graph shows the effect of two different antibiotics on the growth of a population of bacteria.



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3. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

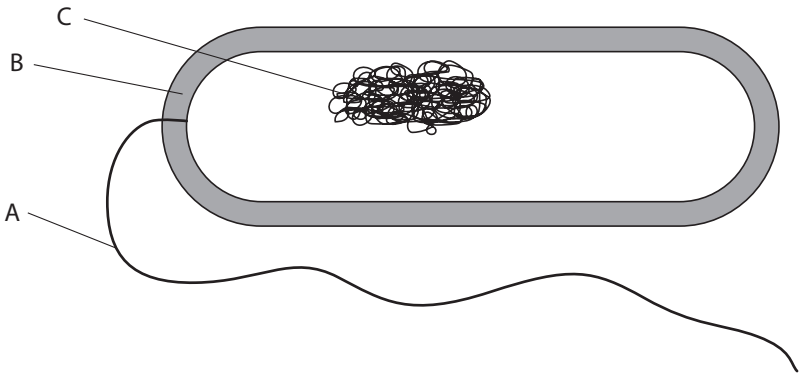
4. Question 8 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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8 The diagram shows a bacterial cell.



(a) (i) Name the parts labelled A, B and C.

(3)

A .....

B .....

C .....

(ii) Some types of bacteria cause disease.

Give the name of organisms that cause disease.

(1)

.....

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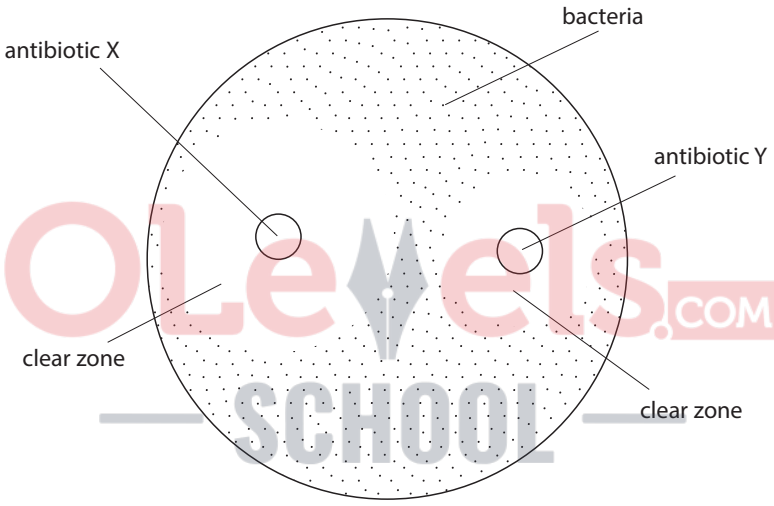
4. Question 8 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

(b) A student uses this method to investigate whether antibiotic X is more effective than antibiotic Y.

- grow bacteria on nutrient agar in a Petri dish
- soak one disc of filter paper in antibiotic X and another disc in antibiotic Y
- place each disc on the agar in the Petri dish
- seal the Petri dish and leave for three days



(i) Describe what is meant by the term **antibiotic**.

(2)

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4. Question 8 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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(ii) Explain which antibiotic, X or Y, is more effective against the bacteria in the Petri dish.

(3)

(iii) Explain a suitable control that could be used in this investigation.

(3)

(Total for Question 8 = 12 marks)

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4. Question 8 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

malaria

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5. Question 4 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

5. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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6. Question 6 · part a.iv · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iv only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	**MRSA**	***S. aureus***	**MRSA**
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

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6. Question 6 · part a.iv · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iv only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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.....

6. Question 6 · part a.iv · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.iv only.

(b) (i) An infection caused by *S. aureus* is usually harmless and can be treated using antibiotics.

State one source of an antibiotic.

(1)

(ii) Describe an aseptic method that a scientist could use to find out whether an infection is caused by *S. aureus* or by MRSA.

(6)

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(Total for Question 6 = 15 marks)

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6. Question 6 · part a.iv · 2 marks · Working and answer

Working and answer

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7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

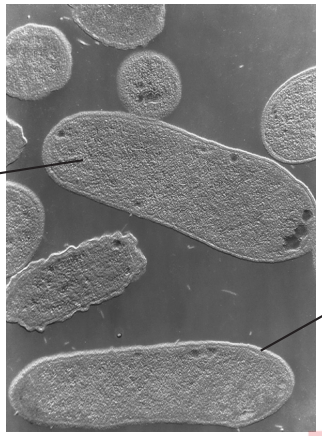
Further practice · Attempt part c.ii only.

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4 Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.



(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

(1)

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

(2)

actual length = ... µm

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7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

(1)

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

(3)

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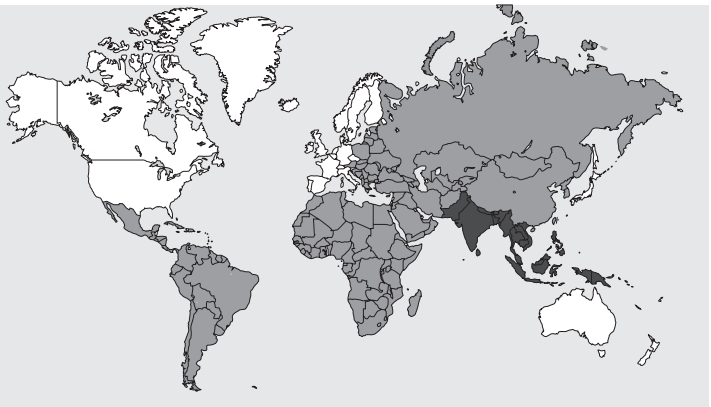
7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

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(c) The diagram shows the levels of typhoid across the world.



(i) Give reasons for the global distribution of typhoid.

(2)

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7. Question 4 · part c.ii · 5 marks

January 2023 Paper 1

Further practice · Attempt part c.ii only.

(ii) Some strains of bacteria that cause typhoid are resistant to antibiotics.

Discuss how resistance to antibiotics develops and the effect this has on the treatment of typhoid.

(5)

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(Total for Question 4 = 14 marks)

7. Question 4 · part c.ii · 5 marks · Working and answer

Working and answer

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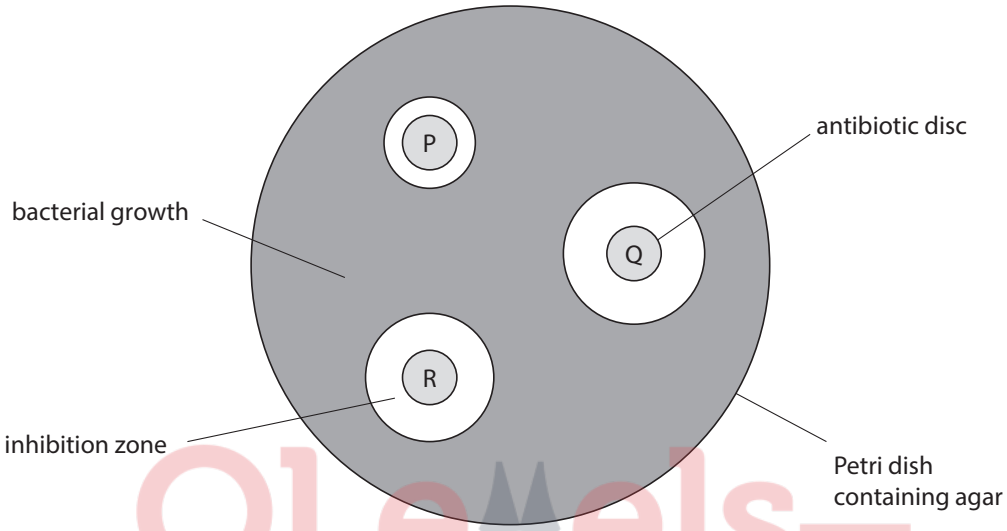
8. Question 5 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

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2 .....

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P 7 4 6 3 0 A 0 1 4 2 4

8. Question 5 · part a.iv · 4 marks

June 2024 Paper 2

Further practice · Attempt part a.iv only.

(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

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(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

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8. Question 5 · part a.iv · 4 marks · Working and answer

Working and answer

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9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

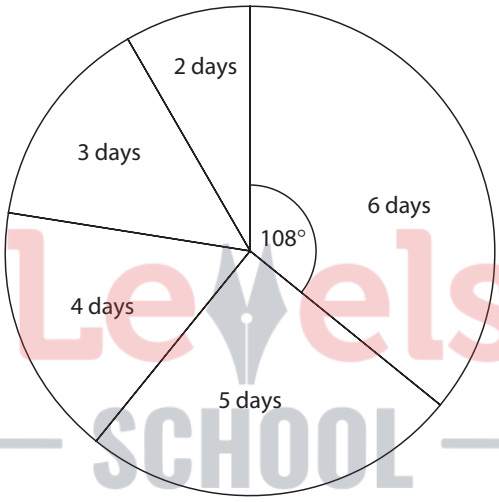
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(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.



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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

(3)

number of types of antibiotic = ...

P 7 4 6 3 2 A 0 2 3 2 4

Turn over ▶

9. Question 8 · part b.iii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(ii) State a reason why it is important to wear gloves and a face mask when investigating bacteria.

(1)

(iii) Describe how the use of antibiotics can cause antibiotic-resistant strains of bacteria.

(2)

(Total for Question 8 = 15 marks)

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Working and answer

Practice workbook · Use aseptic technique to test antibacterial agents on agar

Petri-dish investigations of antibiotics, cleaners or disinfectants: aseptic transfer, contamination control, culture safety and judging effectiveness by clear zones or colonies.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-ASEPTIC-AGAR-PLATE-TEST

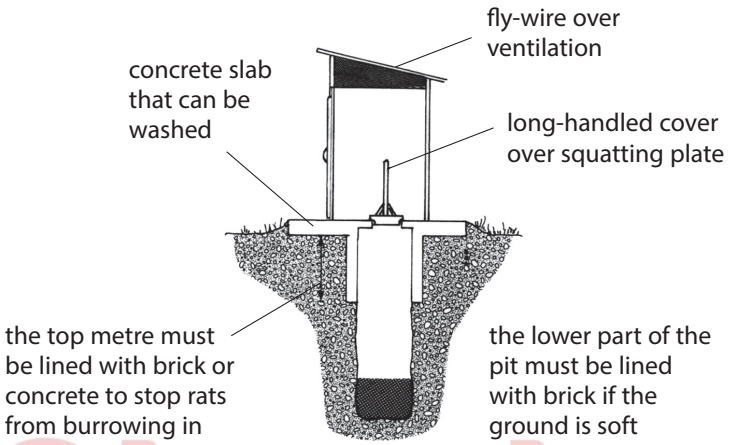
1. Question 5 · part b · 7 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

5 Pit latrines can be used to get rid of urine and faeces. They can be used where the soil is permeable to water and are always placed downhill from wells.

The diagram shows a pit latrine.



(a) (i) Explain why a pit latrine should be placed downhill from a well. **(2)**

(ii) Give the reason why pit latrines must be placed in soil that is permeable to water. **(1)**

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P 7 2 5 8 5 A 0 1 2 2 0

1. Question 5 · part b · 7 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

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(iii) Explain why, when a pit latrine is not in use, the latrine should have a cover and why the cover should have a long handle.

(5)

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1. Question 5 · part b · 7 marks

June 2023 Paper 1

Worked-example retry · Attempt part b only.

(b) The concrete slab shown in the diagram is washed regularly with an antibacterial solution.

A scientist wants to investigate two antibacterial solutions to see which solution is more effective.

Design an investigation the scientist could use to find out which of the two antibacterial solutions is more effective.

Include experimental details in your answer and write in full sentences.

(7)

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(Total for Question 5 = 15 marks)

1. Question 5 · part b · 7 marks · Working and answer

Working and answer

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2. Question 4 · part b.i · 2 marks

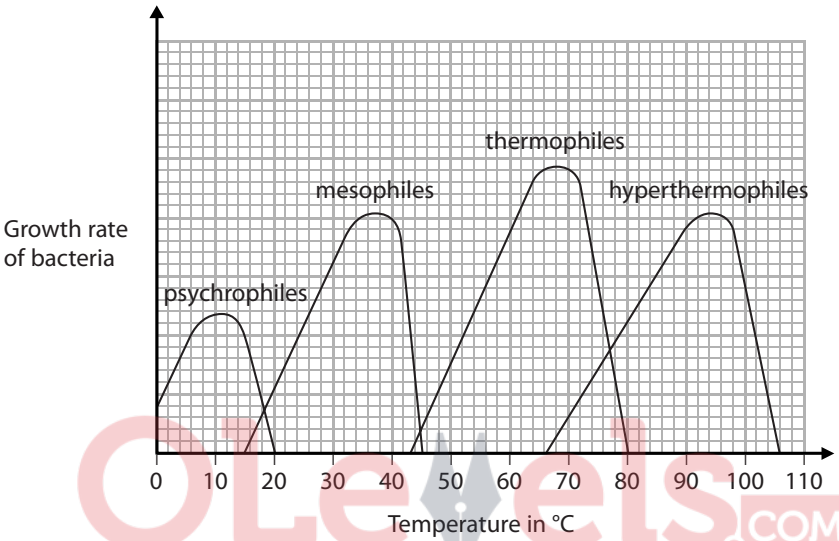
June 2019 Paper 1

Further practice · Attempt part b.i only.

4 (a) Some bacteria in food can affect the digestive system and cause poor health.

Storing food at the correct temperature can reduce the growth of these bacteria.

The graph shows the growth rate of four groups of bacteria at different temperatures.



(i) Food kept in a fridge shows signs of contamination by bacteria.

Which group of bacteria is most likely to have caused this contamination?

(1)

- ☒ **A** psychrophiles
- ☒ **B** mesophiles
- ☒ **C** thermophiles
- ☒ **D** hyperthermophiles

(ii) Explain how cooking food at 65 °C will affect thermophiles.

(2)

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2. Question 4 · part b.i · 2 marks

June 2019 Paper 1

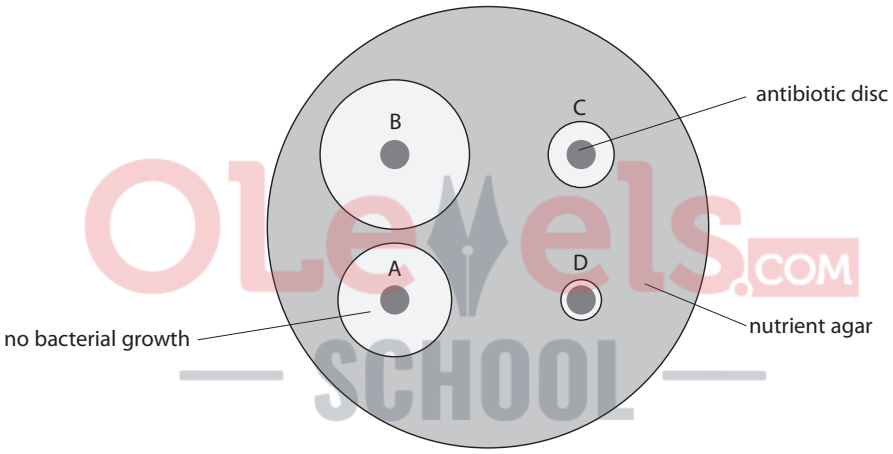
Further practice · Attempt part b.i only.

(b) A student investigates the effect of different antibiotics on the growth of bacteria.

She uses this method.

- streak bacteria onto nutrient agar in a Petri dish
- place discs of different antibiotics, A, B, C and D, onto the nutrient agar
- incubate the Petri dish in a warm oven for one week

The diagram shows the results after one week.



(i) Explain how the student could determine the effectiveness of each antibiotic.

(2)

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2. Question 4 · part b.i · 2 marks

June 2019 Paper 1

Further practice · Attempt part b.i only.

(ii) Explain safety precautions the student should take in this investigation.

(6)

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(Total for Question 4 = 11 marks)

2. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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3. Question 4 · part b.ii · 6 marks

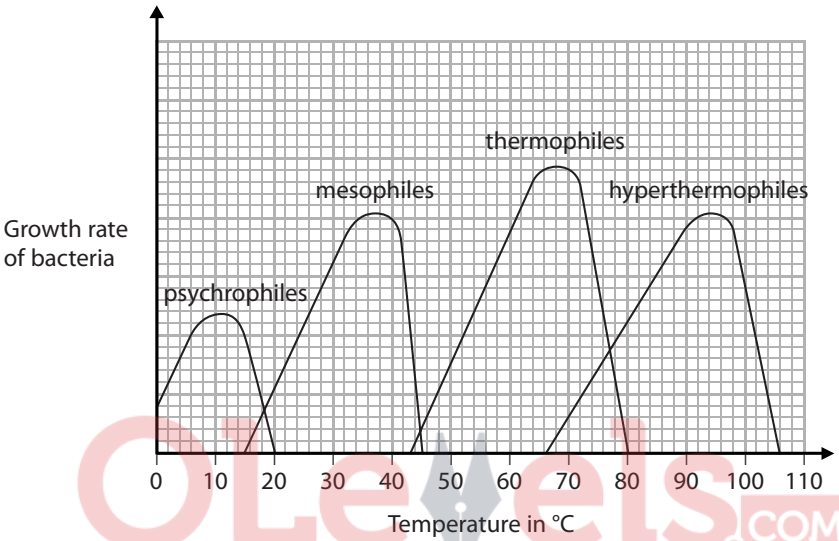
June 2019 Paper 1

Further practice · Attempt part b.ii only.

4 (a) Some bacteria in food can affect the digestive system and cause poor health.

Storing food at the correct temperature can reduce the growth of these bacteria.

The graph shows the growth rate of four groups of bacteria at different temperatures.



(i) Food kept in a fridge shows signs of contamination by bacteria.

Which group of bacteria is most likely to have caused this contamination?

(1)

- ☒ **A** psychrophiles
- ☒ **B** mesophiles
- ☒ **C** thermophiles
- ☒ **D** hyperthermophiles

(ii) Explain how cooking food at 65 °C will affect thermophiles.

(2)

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3. Question 4 · part b.ii · 6 marks

June 2019 Paper 1

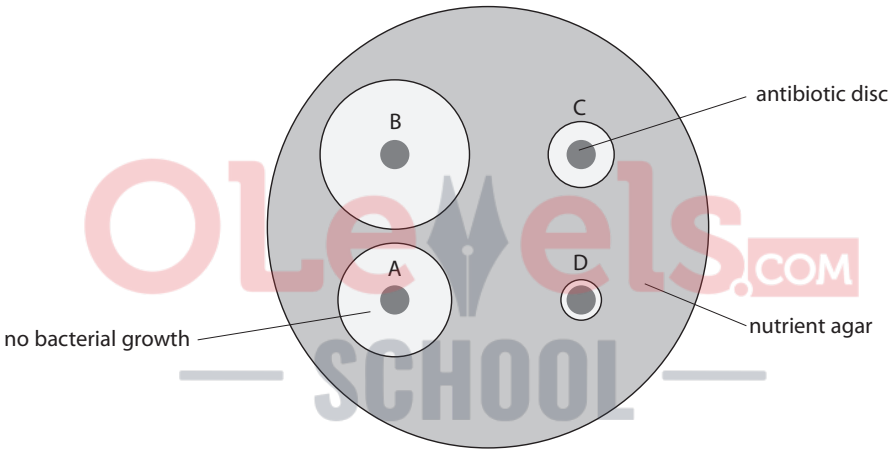
Further practice · Attempt part b.ii only.

(b) A student investigates the effect of different antibiotics on the growth of bacteria.

She uses this method.

- streak bacteria onto nutrient agar in a Petri dish
- place discs of different antibiotics, A, B, C and D, onto the nutrient agar
- incubate the Petri dish in a warm oven for one week

The diagram shows the results after one week.



(i) Explain how the student could determine the effectiveness of each antibiotic.

(2)

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3. Question 4 · part b.ii · 6 marks

June 2019 Paper 1

Further practice · Attempt part b.ii only.

(ii) Explain safety precautions the student should take in this investigation.

(6)

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(Total for Question 4 = 11 marks)

3. Question 4 · part b.ii · 6 marks · Working and answer

Working and answer

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4. Question 6 · part b.ii · 6 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	**MRSA**	***S. aureus***	**MRSA**
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

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P 5 9 9 1 0 A 0 1 4 2 4

4. Question 6 · part b.ii · 6 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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4. Question 6 · part b.ii · 6 marks

January 2020 Paper 2

Further practice · Attempt part b.ii only.

(b) (i) An infection caused by *S. aureus* is usually harmless and can be treated using antibiotics.

State one source of an antibiotic.

(1)

(ii) Describe an aseptic method that a scientist could use to find out whether an infection is caused by *S. aureus* or by MRSA.

(6)

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(Total for Question 6 = 15 marks)

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4. Question 6 · part b.ii · 6 marks · Working and answer

Working and answer

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5. Question 7 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

7 Garlic plants contain a substance called allicin. This substance is thought to have antibacterial properties.

A student investigates the antibacterial effect of garlic.

This is the student's method.

- label three test tubes A, B and C
- add 10 cm^3 of milk into each of the tubes
- add a sample of the bacterium *E. coli* into tubes B and C using aseptic techniques
- crush the garlic in water to form garlic juice
- add 1 cm^3 of garlic juice to tube C
- shake all three tubes and leave to incubate for two hours
- place a sample from tube A into the centre of Petri dish A, from tube B into the centre of Petri dish B and from tube C into the centre of Petri dish C
- incubate the three Petri dishes for five days
- each day measure the diameter of any bacterial growth that has occurred

The table shows the student's results.

Petri dish	Diameter of bacterial growth in mm				
	day 1	day 2	day 3	day 4	day 5
A	0	1.7	3.0	4.6	7.1
B	0	4.2	8.4	15.1	36.5
C	0	0	0	0	0

(a) (i) State three control variables in the student's investigation.

(3)

1 .....

.....

2 .....

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3 .....

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5. Question 7 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

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(ii) State the dependent variable in the student's investigation.

(1)

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(b) (i) Calculate the mean daily increase in the diameter of the bacterial growth for dish A and dish B.

(2)

mean daily increase for dish A ..... mm/day

mean daily increase for dish B ..... mm/day

(ii) Explain the difference between the mean daily increases in the diameter of the bacterial growth for dish A and dish B.

(4)

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(c) Explain what is meant by an aseptic techniques for the transfer of bacteria.

(2)

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5. Question 7 · part c · 2 marks

June 2021 Paper 2

Further practice · Attempt part c only.

(d) Suggest what further tests the student should do to show that garlic can be used as an effective general antibacterial agent.

(2)

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(Total for Question 7 = 14 marks)

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Working and answer

6. Question 6 · part b · 2 marks

June 2022 Paper 2

Further practice · Attempt part b only.

6 A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

- add bacterial culture to sterile agar in a Petri dish using an inoculating loop
- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

(1)

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

(2)

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6. Question 6 · part b · 2 marks

June 2022 Paper 2

Further practice · Attempt part b only.

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(c) Explain why it is important that the Petri dish is incubated at a temperature no higher than 25 °C.

(2)

(d) Give a reason why the filter paper disc soaked in distilled water is included in the investigation.

(1)

(Total for Question 6 = 6 marks)

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Working and answer

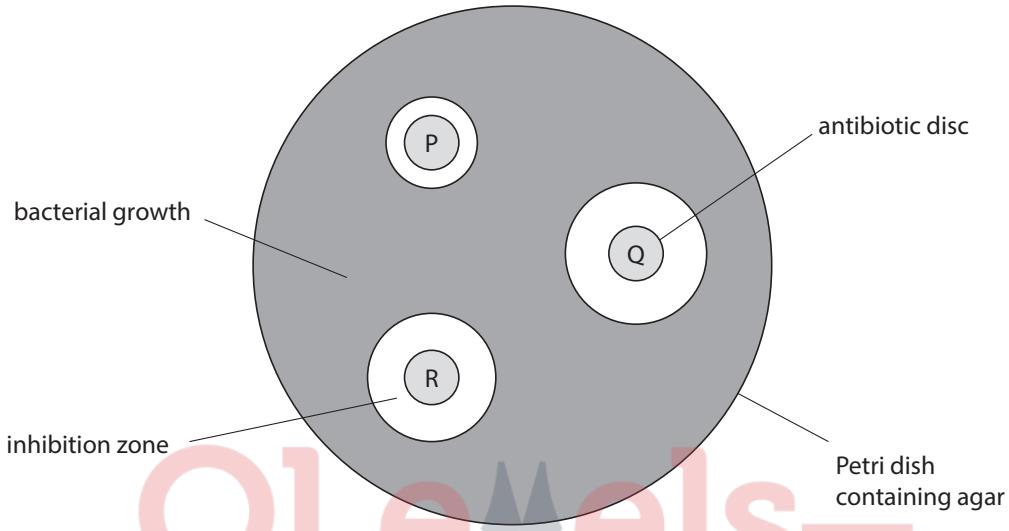
7. Question 5 · part a.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

5 (a) A student investigates the action of three different antibiotics, P, Q and R, on the growth of bacteria.

The diagram shows the results of the investigation.



(i) Which is the independent variable in this investigation?

(1)

- ☒ **A** the mass of bacteria added
- ☒ **B** the size of the inhibition zone
- ☒ **C** the type of antibiotic added
- ☒ **D** the mass of agar in the Petri dish

(ii) Give two procedures that the student should use to reduce the risk of contamination by unwanted bacteria.

(2)

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P 7 4 6 3 0 A 0 1 4 2 4

7. Question 5 · part a.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part a.ii only.

(iii) Explain which is the most effective antibiotic.

(2)

(iv) The student concludes the bacteria are resistant to antibiotic P.

Explain how antibiotic resistance occurs in bacteria.

(4)

(b) Decomposers are microorganisms such as bacteria and fungi that do not cause disease.

Describe the role of these microorganisms in the environment.

(2)

(Total for Question 5 = 11 marks)

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7. Question 5 · part a.ii · 2 marks · Working and answer

Working and answer

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8. Question 6 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

6 A student uses this method to investigate the effect of different antibacterial cleaners on the growth of bacteria.

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- using a pipette, place two drops of an antibacterial cleaner onto a filter paper disc
- repeat using two other filter paper discs, each one soaked with a different antibacterial cleaner
- soak one filter paper disc in distilled water
- place the four filter paper discs onto the sterile agar
- place a lid on the Petri dish
- place the dish into an incubator at 25 °C for 5 days

(a) State the independent variable in this investigation.

(1)

.....

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(b) Give two improvements to this method that will reduce the risk of introducing unwanted microorganisms into the Petri dish.

(2)

1 .....

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2 .....

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8. Question 6 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(c) Explain why it is important that the Petri dish is incubated at a temperature no higher than 25 °C.

(2)

(d) Give a reason why the filter paper disc soaked in distilled water is included in the investigation.

(1)

(Total for Question 6 = 6 marks)

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Working and answer